

Beyond Getting Work Done: A Landscape Review of Semantic Scaffolding for Learning with AI

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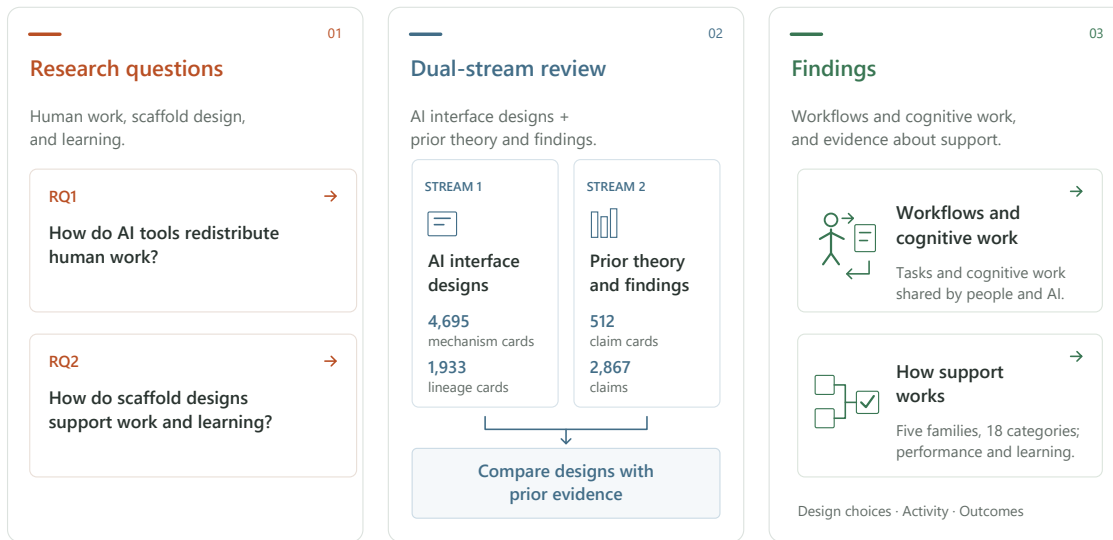


Fig. 1. **The review at a glance.** Two research questions guide a dual-stream review of AI interface designs and prior theory and findings. The findings examine how work changes and how support is designed and evaluated. Card counts cover the selected AI papers in Stream 1 and retained sources in Stream 2.

ABSTRACT

AI tools now help people write, program, analyze data, and design, and many also aim to help people learn through that work. What such a tool gives a person to think with varies: a generated explanation, criteria for judging alternatives, an editable plan, a record of earlier choices. We call these designed structures with task meaning *semantic scaffolding*. We reviewed 4,638 AI-involved papers from 40 venues (2019–2026), together with theories and empirical studies of how support works, and synthesized a design space of 5 mechanism families and 18 categories. Compared with papers on earlier AI, generative-AI papers more often concern producing artifacts and more often give people control over what the system does, after adjusting for venue and year. What a person can edit matters: the task specification, the AI’s interpretation of it, the criteria, or the result. Designs also differ in whether people construct a representation, reason with supplied material, or judge proposed results. Studies that measure learning report gains and costs together: CodeVoyager raised closed-book comprehension without a completion advantage, and required reflection cut completed practice from 3.4 to 2.0 problems in a fixed time. The design space gives researchers named alternatives to build and test.

CCS Concepts: • **Human-centered computing** → **Empirical studies in interaction design**; *Graphical user interfaces*; *Collaborative interaction*; Natural language interfaces.

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1 INTRODUCTION

AI tools help people write, program, analyze information, and make decisions. Increasingly, they supply not just an answer but material people can work with: alternative passages, intermediate plans, explanations, and visual representations [59, 68, 146]. Interfaces let people inspect, steer, and revise this material as a task develops. These possibilities change the division of work between people and tools. Designing assistance therefore involves deciding both what AI should produce and what people should continue to do, understand, and learn.

Help with the current task does not necessarily create opportunities to develop judgment or skill. The assistance dilemma captures this tension: support can enable progress while taking over an activity the learner needs to practice [69]. Learning depends on how people engage with the material, not simply on how much interaction an interface requires [28]. A critical review of HCI research also finds that claims about critical thinking often outrun clear definitions and assessments [58]. Our review finds a related gap in evidence over time: among 396 AI-involved sources in the selected outcome appraisal, only 31 assessed capability in a later session. Designing for development requires identifying worthwhile occasions to interpret, judge, and practice during work, then examining what people learn from them.

We use *semantic scaffolding* to describe designed structures with task meaning that a person inspects or acts through. Examples include a concept map, criteria for comparing alternatives, an editable plan, and a history of earlier choices. Such structures have a dual role: they can help people accomplish work and create opportunities to develop the capability to do it. Scaffolding theory explains responsive support that fades as responsibility passes to the learner [133, 143]; cognitive-artifact research explains representations that remain useful as people become more skilled [100, 108]. Generative and malleable interfaces make it possible to create, reshape, and preserve these structures during use [23, 63, 93]. The design challenge is to make support fit both the decision at hand and the person’s development: some assistance can fade, while useful structures persist or evolve. Section 3 develops this account.

We investigate this challenge through an LLM-assisted, dual-stream landscape review. Language models processed the papers into structured evidence cards with supporting passages and page references. Stream 1 screened 35,480 abstract-bearing records from 40 venues spanning 2019–2026, selecting 7,372 for full-text processing. The current card corpus contains 4,638 AI-involved papers. Inclusion requires AI involvement, an implemented or studied structure with task meaning, and a person who inspects or acts through it. Stream 2 starts from 795 candidate works, of which 543 have full-text claim cards, and contributes 512 retained sources on prior theory and empirical findings, represented by 2,867 claims. We connect the designs to human activity, their theoretical rationale, and evidence about performance and learning. We ask:

- **RQ1.** How do AI tools redistribute human work?
- **RQ2.** How do scaffold designs support work and learning?

For RQ1, GenAI papers more often delegate generation to the tool while also supporting people in creating, editing, and repairing artifacts. Judgment remains necessary, and support can change which information reaches learners, collaborators, and teachers. For RQ2, 5 mechanism families and 18 categories organize choices about what people can inspect, specify, compare, and revise. Comparisons show why the activity matters: CodeVoyager improved immediate comprehension without a detected advantage in assisted completion, while required reflection used practice time without

establishing a transfer advantage [7, 66]. These findings motivate support that creates opportunities for meaningful practice and changes as people develop, while retaining resources useful to their work.

This paper contributes:

- a landscape of how AI interfaces distribute work between people and tools, with quantitative technology comparisons and an account of the cognitive work people retain;
- an illustrated taxonomy of semantic-scaffolding mechanisms, connected to workflow activities and concrete examples in a companion gallery;
- a synthesis linking design choices and prior theory to evidence about process, artifact, and learning, and a research agenda for supporting development within everyday tools.

2 BACKGROUND

Research on AI assistance, learning, and interactive representations poses a shared problem: making a task easier changes what people have an opportunity to do and learn. We review the foundations of that problem, the new interface possibilities, and the gap our review addresses.

2.1 Assisting Work and Supporting Learning

Distributed cognition follows work across people, tools, and representations [48]. External resources can become part of how people remember and reason [31, 57]. Offloading also changes what people retain: expected access to information shifts memory toward where to find it [126], while automation can leave operators with the hardest cases and little practice [8]. These accounts motivate examining the activity left to the person alongside the output of the person-tool system.

AI interaction research makes that division of work a design choice. Mixed-initiative interfaces combine automation and direct manipulation [52]; interactive machine learning centers the user-system loop [3]. Explainable-AI frameworks separate understanding, trust, debugging, and decision support [98], while teaming and learning-support research adds capability and transfer [22, 45]. Generative systems expose prompts, plans, alternatives, and evidence for people to inspect and steer [24, 68, 129, 145]. When systems also supply criteria and explanations, people may need to judge the guidance as well as the answer.

Learning research explains why reducing effort is an incomplete design goal. The assistance dilemma asks when giving or withholding help enables learning [69]; ICAP distinguishes the engagement an activity invites [28]. Guidance useful to novices can become redundant with expertise [61], and difficulty during practice can support later retention [15]. People also trade accuracy against effort [107]. Together, these accounts direct attention to which work assistance removes, which work remains worth doing, and how that balance changes as a person develops.

2.2 Interfaces People Can Inspect and Reshape

Recent design spaces make representational choices explicit. Min et al. separate an abstraction's schema from its representation and examine granularity, presentation, and transformation [94]. Kim et al. organize how generative interfaces elicit preferences along six axes, including when a request appears and which part of an artifact it concerns [64]. These accounts show how interface structure can help people express and work with task knowledge.

People have long adapted technology to their practices. Mackay describes users repurposing email filters from sorting messages before reading to filing them afterwards, leading developers to support both uses [88]. Interfaces can also

adapt controls to a person's abilities [43, 141]. Generative systems extend what can be constructed during use [23]: Google's Generative UI work creates interactive pages and simulations in response to a prompt [77, 78]. Malleable overview-detail interfaces let people change the content, composition, and layout of views [93]; Meridian provides a specification and tools for building those transformations [95]. Maru preserves a user's information-architecture rules across generation [63]. Generating, reshaping, and preserving support offer different ways an interface might respond to changing work and developing expertise.

2.3 What Prior Reviews Establish and Leave Open

Scaffolding syntheses connect contingent support and learner activity to learning [11, 133]. Self-explanation research identifies practice exposure and redundant prompting as consequential conditions [115]. Krakauer distinguishes artifacts whose advantage remains only during use from those that leave capability behind [70]. Inie et al.'s critical scoping review examines how HCI research about AI defines and assesses critical thinking [58]. This work supplies both substantive learning principles and reasons to examine whether a study's measure tests its proposed benefit.

HCI landscapes offer complementary ways to compare designs. Creativity support reviews relate tools to practice and evaluation [40], organize traces of creative activity [47], and examine generative tools in artists' work [68]. Design spaces organize recurring choices and their trade-offs [89]; their visual form shapes what designers explore [14, 75]. Human-AI guidelines already address intelligibility and correction [4]. Citeology, Royal Constellations, and HCI Research Trails make relationships among cited works navigable [19, 92, 97].

Our review connects these levels: a concrete interface choice, the activity it asks of a person, its theoretical rationale, and the outcome a study assesses. To compare an explanation, an editable plan, and a shared view across domains, we need an account of what they provide in common and how their support differs. The next section develops semantic scaffolding as that account.

3 WHAT IS SEMANTIC SCAFFOLDING?

We use *semantic scaffolding* to describe designed structures with task meaning that a person inspects or acts through. The question is how a structure brings goals, relations, criteria, or evidence into a person's work. This lets us compare different interface forms through the relationship between their meaning, the person's activity, and the support supplied. Zong et al. use *semantic scaffolding* for an LLM technique that explains domain groupings of data to blind and low-vision readers [155]; our use extends across these different forms of support.

3.1 Making Task Meaning Available

Semantic concerns what the structure means for the task: which concepts are related, why alternatives differ, what a criterion evaluates, or how an intermediate step contributes to a goal. Its form helps determine what a person can do with that meaning. External-representation research shows how the same information supports different operations when arranged differently [119, 150]. People can also change their surroundings to aid reasoning [67]; coordinating several representations becomes part of the work [2].

Graphologue makes concepts and relationships available through a diagram that expands as a person explores [59]. AI Chains exposes connections among a problem, a suggestion, and drafted feedback in intermediate steps that people can inspect and change [146]. Luminare organizes alternatives along dimensions for comparison [129]. What matters in each case is the task information made available and the operations its organization supports.

Affordances and signifiers explain related aspects of an interface: the actions available to someone and the cues that make them discoverable [101]. Semantic scaffolding asks what task meaning a person can inspect or use through those actions. The same feature can serve both roles. HaLLMark’s highlight signals authorship, while its link to a prompt connects a passage to its production history [51]. Codestrates v2 illustrates why the distinction matters: a file-like tree organized code, but its order also governed execution, a consequence one participant discovered during use [18].

We therefore examine a feature through the task meaning it makes available and the activity it makes possible. A label, score, diagram, or control may expose a relationship, criterion, intermediate state, or consequence for a person to inspect or act on; visual complexity and direct editing are unnecessary. A backend improvement alone supplies no such relationship to human activity. The review’s AI involvement and implementation criteria specify the corpus in Section 4.1.3.

3.2 Supporting Work and Developing Capability

A tool can perform an operation, provide a representation for reasoning, or do both. Norman’s cognitive artifacts store, display, or transform information for a task [100]. A checklist, for example, reorganizes work into preparing, consulting, and interpreting a list. An AI-generated comparison can similarly perform the work of organizing options while leaving the person to judge their relevance. The artifact’s contribution to completing a task and the person’s contribution to it are separate questions.

A developmental account asks how support responds to a learner over time. In the original tutoring account, a tutor takes over parts a learner cannot yet manage so the learner can undertake the rest [143]. Developmental scaffolding responds to the learner, reduces assistance as capability grows, and transfers responsibility [133]. It can make work more manageable or direct attention to an important problem the learner must reason through [112, 113]. Support can be distributed across people, artifacts, and tools [10, 108].

These perspectives explain the dual role we examine: a structure can help someone accomplish work and create opportunities to develop through it. Whether it serves the second role depends on the learner and the activity. Pea distinguishes continuing cognitive support from developmental scaffolding that fades [108]; Sherin et al. offer a broader comparative account of scaffolded activity [123]. Puntambekar and Hübscher identify the risk of calling tools scaffolds without responsive help or a transfer of responsibility [111]. We preserve that distinction within our broader meaning-focused account: developmental properties and learning evidence are examined separately from the presence of a structure.

3.3 Support That Fades, Persists, or Evolves

Support can change without disappearing. Consider a comparison table used by a novice to work through options. Prompts to explain each choice might be reduced as the person becomes more capable, while the table remains useful for organizing a larger set of alternatives. Developing capability can involve using a continuing resource more effectively, knowing when to consult it, or becoming less dependent on it. An assessment must specify what the person has learned and what assistance remains; the quality of the supported output alone describes what the person–tool system achieved [100, 108].

Generative and malleable interfaces widen the choices for this evolution. The system can generate new material, a person can reshape a view, and the person’s organizing rules can persist across changes [23, 63, 93]. Co-adaptation also includes people changing their practices and repurposing support [88]. Adapting to a new task and adapting to a learner’s development have different purposes. For learning, the decision is which activity the person should undertake

now and which support helps them undertake it. This connects the review’s two questions: how AI tools distribute work, and how scaffold designs support that work and learning. We record the designed structure, the activity documented in use, and the outcome assessed to examine that relationship.

4 METHODS

We reviewed two bodies of work (Figure 2). Stream 1 is a systematic corpus of AI-involved systems and studies from fixed venues and years; it supplies the designs we map. Stream 2 collects theories and empirical studies of how support works, including studies without AI; we interpret the designs against it. The two are counted separately. Scoping-review guidance shaped selection, extraction, and reporting [6, 76, 131], and earlier HCI landscape reviews shaped the venue frame and synthesis [40, 47].

4.1 Literature Review

4.1.1 Venues. Stream 1 covers 40 venues in HCI, design and creativity, visualization, graphics, learning and education, software engineering, and AI and language, published from 2019 to 2026 (Figure 3). Papers are dated by their version of record; the selection figure, the design space, and the trend figures cover papers published by 16 July 2026. Ineligible publication types were excluded by rule, and citation counts played no part.

4.1.2 Screening. Two language models separately screened the 35,480 records with an abstract; the 2,498 without one were not screened. Their verdicts, a review of their disagreements, and (for the first reading list) PDF availability determined the 7,372 records selected for full-text reading. One language model processed each full text and recorded whether it met the criteria, with supporting passages and page references.

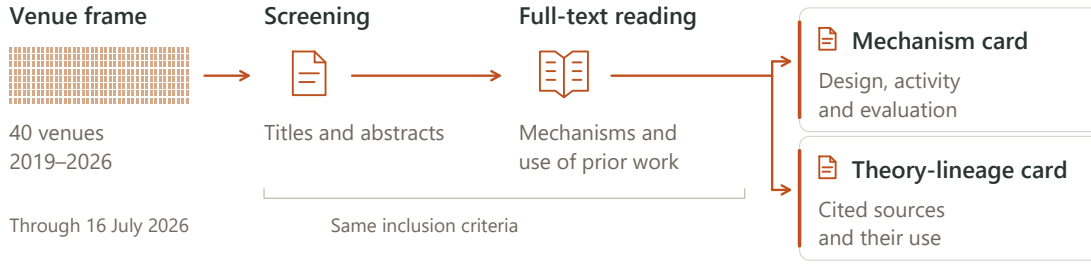
The AI requirement was added partway through full-text reading, so three separate model passes rechecked the 1,154 earlier cards that left the question open. A paper left the corpus only when all three found no AI, with low uncertainty and a quoted passage (492 cards). The corpus holds 4,638 papers, and 1,076 of them form a learning subset: their support has a documented learning purpose, or the study examines its learning consequences. Another 475 papers without AI form a labeled comparison set outside the systematic corpus counts; the text marks any it uses as an example.

4.1.3 Inclusion Criteria. Stream 1 includes a paper when three conditions hold, at screening and at full text (Figure 4): AI is part of the system or the study; the paper implements or studies a designed structure with task meaning; and a person inspects or acts through that structure. AI is part of a paper when an AI system is built, used, evaluated, or simulated (including Wizard-of-Oz systems presented as AI), or when AI is what the study designs, explains, or labels. Rewording a chatbot’s replies (hedges, tone, length) is not a designed structure. Workshop templates, cards, storyboards, and probes used to elicit views count only when the paper builds and evaluates them as its contribution. Reviews, surveys, and meta-analyses go to Stream 2. A design can qualify without a user study; its benefits then count as intended uses, not measured effects. Learning purpose, measured outcomes, adaptation, fading, and the transfer of responsibility are recorded but do not decide inclusion.

4.1.4 Theories and Empirical Studies. Stream 2 gathers theories, design principles, and empirical findings about how support works. Its sources came from database searches, backward and forward citation snowballing [142], and reviews found in Stream 1, and every source received the same full-text assessment of relevance and support. Of 795 candidate works, language models processed 543 full texts into claim cards; the model-assisted assessment excluded 31 for recorded reasons, such as falling outside the review questions or appearing in a questionable venue. The analysis uses

Literature selection and evidence extraction

Stream 1 · AI-involved designs



Stream 2 · Prior theory and findings

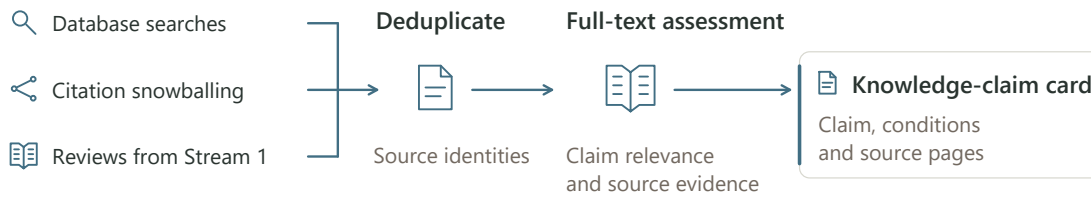


Fig. 2. **Literature selection and evidence extraction.** Stream 1 runs from a venue frame through screening and full-text reading to mechanism and theory-lineage cards. Stream 2 combines three discovery routes, removes duplicates, and records each source's claims on a knowledge-claim card.

the remaining 512 sources and their 2,867 claims. Books, foundational theory, null findings, and studies without AI are eligible.

4.2 Analysis

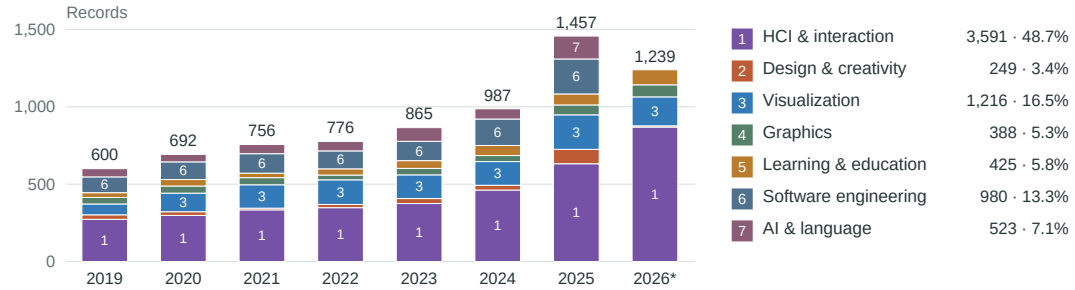
RQ1 asks how AI tools redistribute human work. We examine the activities interfaces support, the operations assigned to tools and people, and how these patterns vary by technology, year, and context. RQ2 asks how scaffold designs support work and learning. We compare mechanism choices, their use within workflows, their theoretical rationale, and the process, artifact, and learning outcomes assessed by studies.

4.2.1 What We Recorded. A language model processed each full text into a structured evidence card, quoting the passages and pages behind its entries.¹ Scientific-document benchmarks and expert-assisted literature mining offer precedents for such passage-linked records [134, 139]. The Stream 1 unit is a *mechanism*, an operation involving a designed structure with task meaning: AI Chains' workflow overview and its step editor are two mechanisms [146]. For each mechanism, the card records what the interface represents, how and when it presents that meaning, and what people can do. It also records how the work changed, what the tool took over and what the person kept, and how the design was evaluated: comparison, measures, duration, participants, and results. A second card records how the paper

¹Evidence cards: mechanism, theory-lineage, and knowledge-claim cards .

Selected records by year and venue group

7,372 records selected for full-text reading; venue groups describe disciplines, not application domains.



Each venue: selected records and share of the 7,372 selections

1 CHI	1,548 · 21.0%	1 ECSCW	10 · 0.1%	2 Design Studies	13 · 0.2%	5 AIED	58 · 0.8%
1 UIST	306 · 4.2%	1 GROUP	6 · 0.1%	2 IDETC	151 · 2.0%	5 IJAIED	37 · 0.5%
1 IUI	359 · 4.9%	1 ASSETS	84 · 1.1%	3 VIS	394 · 5.3%	5 EDM	103 · 1.4%
1 DIS	232 · 3.1%	1 HRI	187 · 2.5%	3 EuroVis	111 · 1.5%	6 CHASE	55 · 0.7%
1 CSCW	258 · 3.5%	1 IVA	82 · 1.1%	3 TVCG	635 · 8.6%	6 ICSE	321 · 4.4%
1 C&C	67 · 0.9%	1 CHIIR	56 · 0.8%	3 VL/HCC	76 · 1.0%	6 FSE	238 · 3.2%
1 TOCHI	45 · 0.6%	1 HCOMP	36 · 0.5%	4 SIGGRAPH	295 · 4.0%	6 ASE	366 · 5.0%
1 TiIS	102 · 1.4%	1 IJHCS	127 · 1.7%	4 SIGGRAPH Asia	93 · 1.3%	7 ACL	191 · 2.6%
1 EICS	33 · 0.4%	2 ICCS	53 · 0.7%	5 L@S	41 · 0.6%	7 EMNLP	255 · 3.5%
1 HAI	53 · 0.7%	2 Design Science	32 · 0.4%	5 LAK	186 · 2.5%	7 RecSys	77 · 1.0%

*2026 is partial: publication cutoff 16 July. Zero selections can reflect unavailable proceedings.

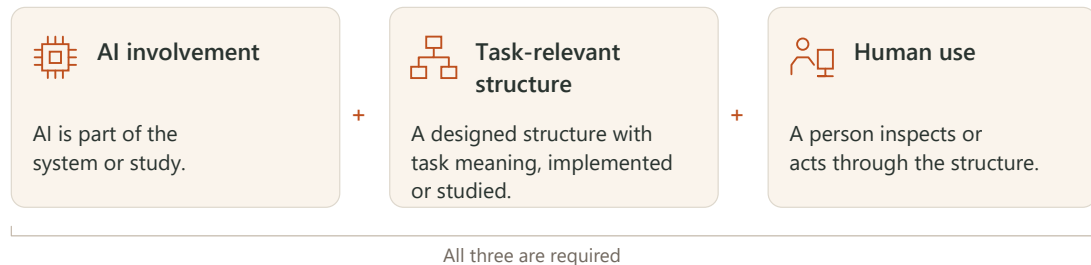
Fig. 3. **Records selected for full-text reading, by publication year and venue group.** The lower panel gives each venue's count and share of the 7,372 selections. The 2026 bar covers papers published by 16 July.

Inclusion criteria

Stream 1

Title/abstract screening

Full-text reading



Stream 2 · Prior theory and findings

Relevant, transferable claim

Full-text support and conditions

Fig. 4. **Inclusion criteria.** Stream 1 requires all three conditions at screening and full text. Stream 2 assesses whether a claim is relevant and supported by its full text. Learning purpose and measured outcomes are recorded during extraction.

uses each cited source: to frame a problem, explain a mechanism, motivate the design, inform a measure, critique an approach, or in passing. The Stream 2 unit is a *knowledge claim*, a finding or theoretical argument about how support works, for whom, and when; each card holds one to eight claims with their evidence, conditions, and limits. Missing information is recorded as not reported or unclear, and author claims, measured results, and reviewer inferences stay apart (fields in Appendix A).

4.2.2 Building the Design Space. Closed fields take predefined values, such as whether learning was measured and for how long; open fields describe mechanisms and human work in each paper's terms. We merged the 142,877 open-field observations into 1,290 codes and grouped these into 223 categories and 92 families across 18 fields (Appendix A.3). Each grouping links to its observations, so the grouping and the assignments can be checked separately [54]. The mechanism field's hierarchy is the design space: 5 families and 18 categories (Section 6; codebook in Appendix B). For the four-function view (Figure 12), we assigned each of the 110 mechanism codes to the functions it serves; exploring alternatives serves both representing and evaluating.

We built the vocabulary from the cards completed by 28 August 2026 and coded later cards with it. The design-space counts describe the 2,847 papers of that first set still in the corpus (2,891 for the closed fields); the annual and technology figures draw on the 4,133 papers with usable coding (Appendix A.6). A paper counts once per category, however many mechanisms it has. We classified each paper's AI as generative or traditional AI/ML by the technology it used, not its year, and adjusted technology comparisons for year and venue, since venue mix alone can create differences (Appendix D).

4.2.3 Comparing Designs and Studies. We used the codes to find cases and the source passages to compare them. For each question, we compared the task, the representation supplied, the actions available, the reported use, and the assessed consequence, recording contrary and unclear cases beside supporting ones. We chose examples that fit a mechanism, varying domain and year; citation counts mattered only after fit. We retrieved prior findings and counterexamples through the seven fields coded on Stream 2 claims, such as the form of support and a claim's conditions. The lineage cards trace how a cited theory or finding entered a design or a measure (Section 6.5); reference lists and citation counts come from Semantic Scholar (Appendix A.9.2).

In the selected lineage comparisons, we examined the connection between the source's claim, the implemented feature, the activity attributed to the person, and the assessment. This distinguishes a design rationale from evidence that the proposed activity occurred or helped. Unavailable lineage evidence remains unknown, rather than evidence that a design has no theoretical grounding.

For study results, each contrast keeps its population, comparator, outcome, timing, and the assistance available during assessment, and positive, mixed, null, harmful, and unmeasured results stay distinct. A whole-system result is credited to the whole system, and a learning claim counts as tested only when a study measured learning or transfer. We sorted outcomes into process, artifact, and learning (PAL), each assessed by the study or reported by participants (Section 6.6). We did not pool effects across studies (Section 9).

4.2.4 Checking the Coding. Scripts checked the cards' required fields and matched their quotations against the source text, and quotations the text layer could not match were checked against page images; counts and plots are computed from the saved data. To test the open-code grouping, a separate model pass coded 1,571 extracted observations in six fields with the existing codes hidden. Agreement was low (adjusted Rand index .079 to .445), and the blind pass grouped

more coarsely in five of the six fields (Appendix A.4). The mechanism vocabulary gained 12 codes across the 1,754 papers read after it was built (Appendix E).

4.3 AI Use Disclosure

We used language models to screen records, process full texts into evidence cards, code observations from those cards, group codes, develop and critique the cross-case comparisons, write analysis code, and draft and revise text, tables, and figures. Claude Opus 5 and GPT-5.6 Sol screened titles and abstracts. GPT-5.6 Sol, Claude Opus 5, and Claude Opus 5.5 processed full texts into the Stream 1 cards; Claude Opus 5 and GPT-5.6 Sol wrote the Stream 2 cards; Claude Opus 5.5 ran the three-pass AI check. Claude Opus 5 merged labels into codes, and model-assisted review, including Claude Table 5.1, grouped the codes into categories and families. GPT-6 Sol linked card evidence for the cross-case comparisons, and GPT-6 Astra developed and critiqued them. GPT-6 Astra, working through Codex, drafted the merged manuscript, and Claude Opus 5.5 revised it. Prompts and run records are kept. The authors set the research questions, scope, criteria, and card formats, directed the synthesis, and are responsible for its conclusions; people have not yet checked the coding or the cross-case interpretations (Section 9).

5 FINDINGS: WORKFLOWS AND COGNITIVE WORK

AI assistance changes what people work with and what they have to do. We examine three consequences: generation gives people more material to select and revise; judgment still depends on useful ways to check that material; and support can change who has the information needed to help someone else.

5.1 More generated material, more selection and revision

Research on cognitive artifacts distinguishes what a person and tool achieve together from the task left to the person [100]. Creativity-support research likewise examines how tools support activities such as exploring possibilities and developing ideas [40, 68]. The AI designs in our review make this distinction concrete. CorpusStudio retrieves examples and patterns from prior writing for authors to interpret before composing their prose [32]. ABScribe generates alternative passages that authors keep, edit, and compare [114]. The material supplied changes the writer’s starting point and the choices available next.

GenAI moves generation to the tool. In 2,119 papers from 2019–2025 with coded delegated work, the tool generates the artifact in 19.3% of GenAI and 1.9% of traditional AI/ML papers (+16.6 points after adjusting for year and venue; 95% CI 12.7 to 20.5), and it more often generates responses and revisions (+9.4). Analysis moves the other way: the tool less often calculates and models (−11.5), evaluates and classifies (−9.7), or compares and finds patterns (−6.7) (Appendix D.2).

The human activity codes show a corresponding emphasis on working with artifacts (Figure 5). Among 2,426 technology-classified papers with activity coding, creating, editing, or repairing artifacts appears in 359 of 1,213 GenAI papers (29.6%) and 183 of 1,213 traditional AI/ML papers (15.1%). Specifying or refining system behavior appears in 37.6% and 30.0%, respectively; comparing alternatives or testing scenarios appears in 11.6% and 20.0%. These pooled shares describe the activities supported by designs across 2019–2026; the delegated-work comparison above separately adjusts for year and venue in 2019–2025. Across these separately coded subsets, GenAI papers more often describe tool generation and human creation, editing, or repair. The group-level patterns motivate examining how designs connect those operations within a task.

The technology contrast concerns supported operations. The separate domain analysis finds that none of the 16 trends meets the corrected significance threshold after accounting for venue between 2019 and 2025 (Table 12). The

Workflow activities: volume and annual share

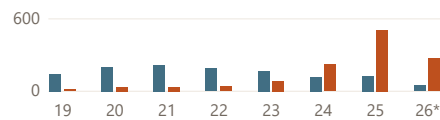
2,426 coded papers with assigned AI technology - categories overlap



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 5. Workflow activities by AI technology. Bars show counts across all years; annual lines show shares within each technology group. Denominators count papers with usable activity codes and a known technology. Activities overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue.

quantitative pool contains 4,133 AI-involved papers; each comparison uses those with the required codes. Technology classification follows the system used, including earlier generators and generative hybrids. GenAI accounts for 9.2% of

What does the person do?

Twelve activities that interfaces can support

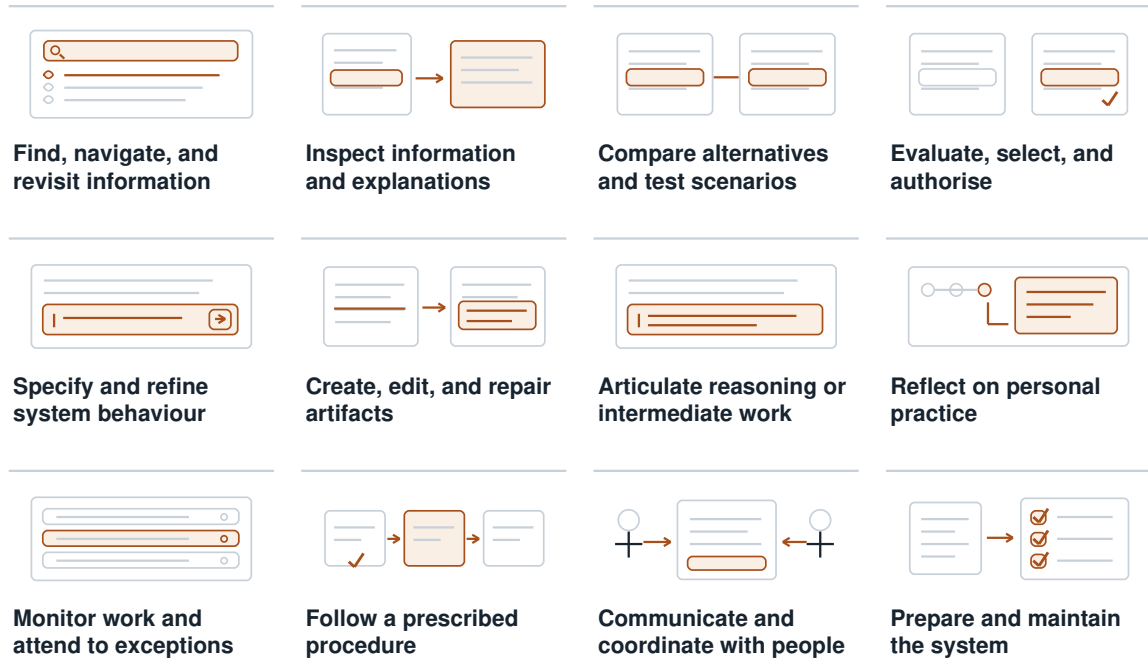


Fig. 6. What does the person do? Twelve activity groups organize the workflow codes. Each schematic illustrates an action through an interface: finding, inspecting, comparing, judging, directing, creating, articulating, reflecting, monitoring, following a procedure, coordinating, or preparing a system. These activities can occur together; the grid does not imply a sequence. Allocation, supplied support, and measured outcomes are coded separately.

classified papers in 2019 and 80.7% in 2025 (Figure 56). Appendix D gives the annual counts and domain distributions; 2026 covers publication through 16 July.

Figure 6 organizes the activities people can carry out with these designs. Activities can recur or combine within a task: an author may inspect examples, specify an intended change, edit a passage, and then reconsider the examples. The map describes kinds of activity; coverage of several kinds alone does not establish support for a complete workflow.

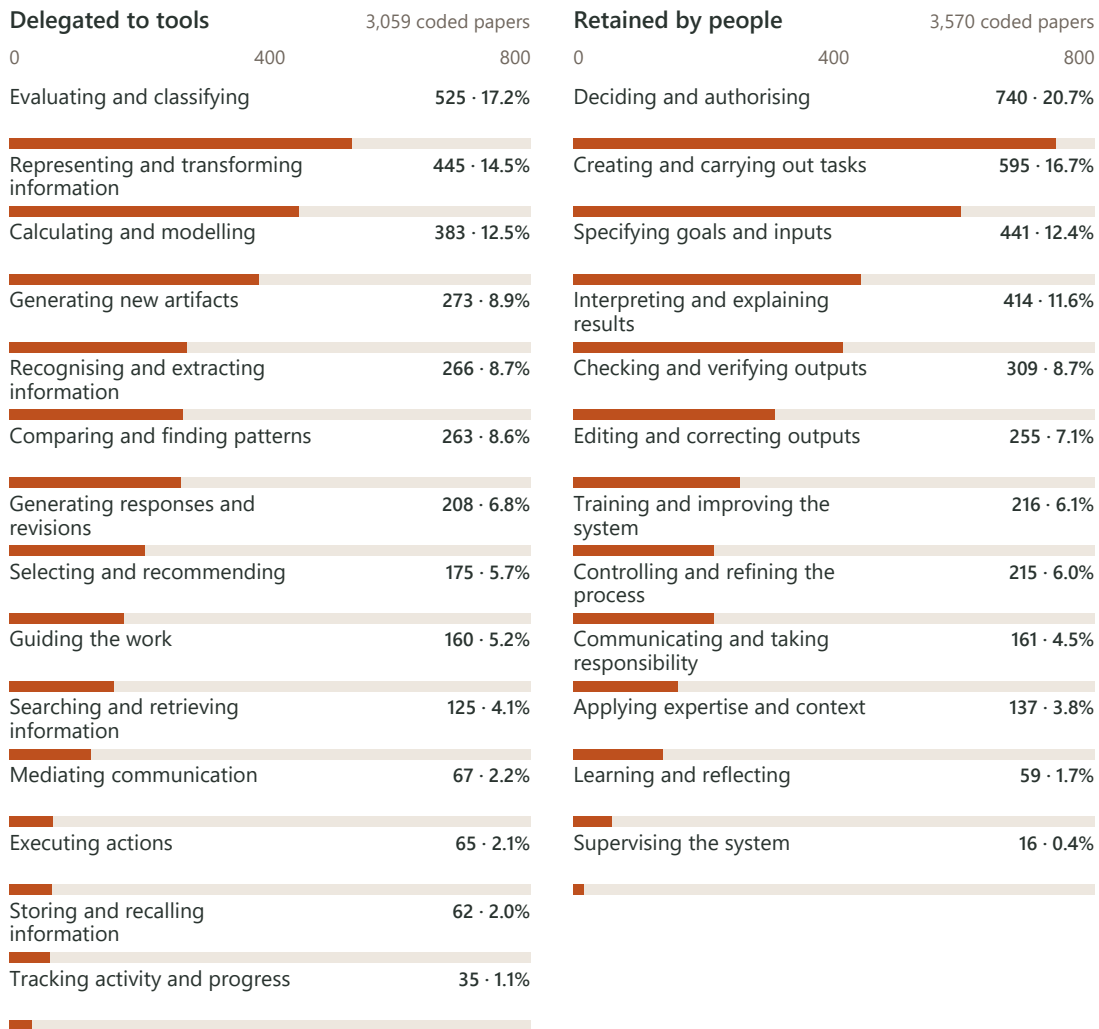
5.2 Judgment needs a basis for checking

Generating material leaves consequential work to the person who will use it. The retained-role codes locate that work: deciding or authorizing appears in 740 of 3,570 papers (20.7%), specifying goals and inputs in 441 (12.4%), and checking or verifying in 309 (8.7%) (Figure 7). The tool and person panels have separate coding denominators and overlapping roles; they describe operations, not amounts of mental effort or improvements in capability.

Figure 8 shows how these operations fit together. A tool can compose material while a person judges its fit, revises it, or checks its evidence. The design question is which information makes that judgment possible.

Work delegated to tools and retained by people

Bar length: number of papers · percentages use each field's coded papers



Counts use all papers coded for each field. Allocation conditions and unspecified work are listed with the data.

Fig. 7. Delegated and retained work, ranked by prevalence. Bar lengths encode paper counts on a common scale; labels also give percentages within each panel's coding denominator. A paper can record several roles. Each panel uses papers with usable codes for that role; technology classification is not required.

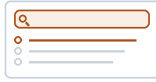
Different references support different checks. Leap exposes values produced when generated code runs [39]; CorpusStudio offers precedents from a writing corpus [32]; ONYX shows how it interpreted an action the user taught it [117]; and Rehearsal simulates how a conversation partner might respond [121]. Execution, precedent, interpretation,

How is the work divided?

The tool supplies an operation; the person still has work to do.

What the tool does

Perception, search and memory



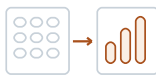
Recognising and extracting information
Searching and retrieving information
Storing and recalling information
Tracking activity and progress

Representation and translation



Representing and transforming information
Comparing and finding patterns

Computation and judgement



Calculating and modelling
Evaluating and classifying

Content generation



Generating new artifacts
Generating responses and revisions

Decision, guidance and action



Guiding the work
Executing actions
Selecting and recommending
Mediating communication

What the person retains

Judgement and decision



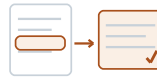
Interpreting and explaining results
Deciding and authorising
Specifying goals and inputs
Applying expertise and context

Doing and revising the work



Creating and carrying out tasks
Editing and correcting outputs
Controlling and refining the process
Training and improving the system

Checking the output



Checking and verifying outputs

Responsibility, social and self-directed work



Communicating and taking responsibility
Learning and reflecting
Supervising the system

Fig. 8. How is the work divided? Five families describe operations supplied by tools; four describe work retained by people. The two columns are independent accounts of a design, not paired transfers. Labels distinguish fourteen tool operations and twelve human activities; learning and reflection are separate from supervising a system. Failed, withheld, absent, and unspecified delegation are recorded separately. Browse examples of tool and human work.

and simulation answer different questions. A simulation can support rehearsal without predicting what a particular person will say; a corpus can supply examples without establishing that an idea is new.

Making a reference available is only part of the work. ONYX’s full interface improved script accuracy, yet users missed or wrongly dismissed 44.8% of its follow-up questions [117]. Leap’s live values did not significantly change validation success compared with a condition in which people could still run the code themselves [39]. Together these cases point to a concrete design problem: help people recognize when a check matters, interpret its information, and act on what it reveals.

Generation can also preserve the activity someone needs to practice. ExPeerience uses AI to create problem contexts, tables, and explanations while leaving construction of the SQL solution to the learner [153]. The placement of assistance matters: generating the exercise and generating its solution leave different opportunities to think and act. Section 6 examines the mechanisms that make these choices possible and the evidence for their consequences.

5.3 Who can see the work changes who can help

Distributed-cognition and scaffolding accounts follow support across people and representations [48, 108]. The reviewed designs show why this matters: the person using an interface, the person whose activity it represents, and the person expected to benefit can be different. A useful handoff carries information that someone else can interpret and respond to.

In entrepreneurship coaching, novices describe a project and select risks; a mentor receives selected and omitted risks with source traces before a meeting [55]. HTCT turns tutor discourse into analytics that a coach selects and discusses with the tutor [17]. One carries the novice’s priorities into a conversation; the other makes teaching activity available for feedback. The studies follow different parts of the response: the entrepreneurship study observed real coaching meetings, whereas HTCT followed tutors’ subsequent practice over seven weeks.

Software feedback can also make a learner’s difficulty less visible to another person. In WriteAid (Stream 2), some recurring questions became less visible to the teacher when pupils sought software feedback, while the teacher gained time to circulate and new peer conversations occurred [99]. Assistance changed both access to information and the opportunities to respond. This suggests examining who no longer sees a question as well as who can now answer it.

The recipient’s response distinguishes a proposed connection from an observed change in practice. eaSEL can share a child’s reflective artifact with a parent and prompt a conversation, but its evaluation did not observe the whole parent–child exchange [122]. M-Powering Teachers, in contrast, provided mentors with feedback on their own talk; randomized access increased their uptake of students’ contributions and reduced their share of talk [33]. The latter establishes a change in teaching practice, while student learning remains a separate question. For shared support, the important sequence is therefore what travels, who receives it, and what they do next. The next section examines the interface mechanisms that shape these possibilities.

6 FINDINGS: MECHANISM TAXONOMY

Mechanisms help people make sense of material, shape and guide action, and coordinate with others. The 5 mechanism families and 18 categories distinguish interface choices within these overlapping concerns. We first map the mechanisms and their prevalence, then examine how they work together through an iterative activity. Comparisons of revision, judgment, and practice show what particular designs make possible. Prior learning research explains some of their rationales; the final subsections examine what their evaluations establish.

6.1 Mechanism families

Prior design-space reviews make interface choices visible: how abstract ideas take concrete form, or how an interface elicits a person’s preferences [64, 94]. Scaffolding research adds the distinction between structuring a task and making important work available for the learner to confront [113]. We use the mechanism taxonomy to connect these design choices to the work people do and the outcomes studies assess.

A mechanism pairs an operation with its form: a constraint can express a goal, an authored rule, or an adjustable threshold. Figure 9 organizes these forms, and Figure 10 compares their prevalence across technologies; a paper can use several. Table 17 counts the papers in each category and how many of them concern or measure learning, and the category maps in Appendix C.1 lead to particular mechanisms and their source papers.

Making sense of the work. The *organization and comparison* family supplies alternatives, relationships, and histories. Graphologue turns text into entities and links [59]; Crystalline gathers options, criteria, and evidence from the pages a developer reads into a comparison table whose cells lead back to their sources [80]. The *explanation and visibility* family exposes sources, rationale, or state. HaLLMark traces copied LLM text to its prompts but relies on writers to mark the passages they consider AI-influenced [51]; automatic provenance captures direct reuse more readily than inspiration.

Shaping and guiding action. The *user control and expression* family lets people state intentions and act on the material: Plume can lock text while it regenerates the explanation around it [79]. The *guidance and feedback* family supplies recommendations, questions, and task structure. OATutor selects practice and reveals hints [106], while WithYou shortens the pauses it inserts into speech as a learner’s shadowing improves [151]; revealing more help and reducing help are different interventions.

Coordinating perspectives. The *collaboration and perspective* family makes contributions and interpretations available to other people: Wikibench supports discussion that revises label definitions and inclusion rules [72]. The concerns combine: a shared comparison can help people understand alternatives, act on them, and discuss their reasons.

To use the taxonomy, start from a decision the person must make. Find the mechanism that supplies information or an action for it, then ask what response it requires and which outcome would show success. A provenance link, for example, gives a basis for checking a claim, but the person still judges whether the source supports the wording. The link shows traceability; accurate checking, or learning to judge sources, needs its own observation.

6.2 Mechanism patterns and combinations

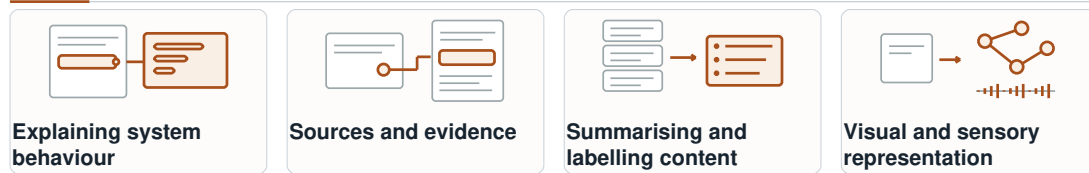
Technology and annual patterns. GenAI control steers generation rather than correcting output. GenAI papers more often describe user control and expression (+13.8 points after adjusting for year and venue; 95% CI 8.9 to 18.7; Appendix D.2). In the 2,532 papers with facet codes, two control facets show a reliable gap, and both put generated candidates before the person: GenAI systems more often generate alternatives to explore (+4.8; 95% CI 3.0 to 6.6) and complete or continue the person’s own content (+3.0; 1.4 to 4.6; Figure 9). The work people keep leans the same way, toward steering the loop (+3.4; 1.2 to 5.6) and away from improving the system (-5.9; -7.7 to -4.0; Figure 8). Acting on the output and stating criteria are not more common: editing machine output (+1.8; -1.4 to 5.1), direct manipulation of a semantic object (+0.9), and specifying criteria and rubrics (-2.8; -4.9 to -0.8). A choice among generated alternatives still needs criteria and contrasts, which few of these interfaces ask people to state (4.4% of GenAI papers). Section 5.2 and Table 6.3 compare designs that supply them.

What does the interface make possible?

Five design families · eighteen categories · 110 retained distinctions

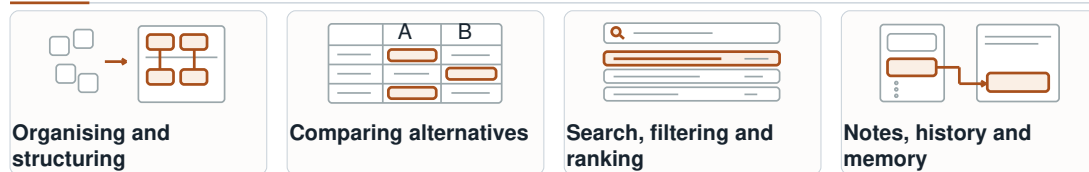
Explanation and visibility

What can a person inspect and understand?



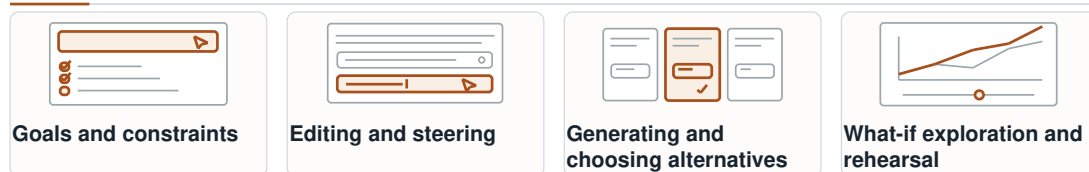
Organisation and comparison

How can a person organise and compare material?



User control and expression

What can a person express, change or try?



Guidance and feedback

How is the next action or reflection supported?



Collaboration and perspective

How can people share work and perspectives?

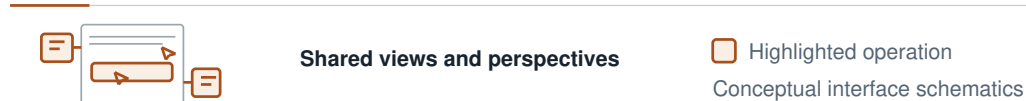


Fig. 9. Five design families organise eighteen categories of semantic scaffolding. Each small interface schematic illustrates a category, with its operation highlighted in orange. A mechanism can combine operations across these branches; layout does not encode prevalence or effectiveness. Complete category maps and source-paper illustrations appear in Appendix C.1. Explore the design atlas; search all 110 distinctions.

GenAI systems rank items less often. Organization and comparison is less common in GenAI papers (Figure 10). Its largest facet gap is ranked or prioritized surfacing, which orders items so the important ones come first (3.2% of

GenAI papers more often offer user control

GenAI versus traditional AI/ML, accounting for publication year and venue

2,453 papers with family coding and classified technology · 2019–2025

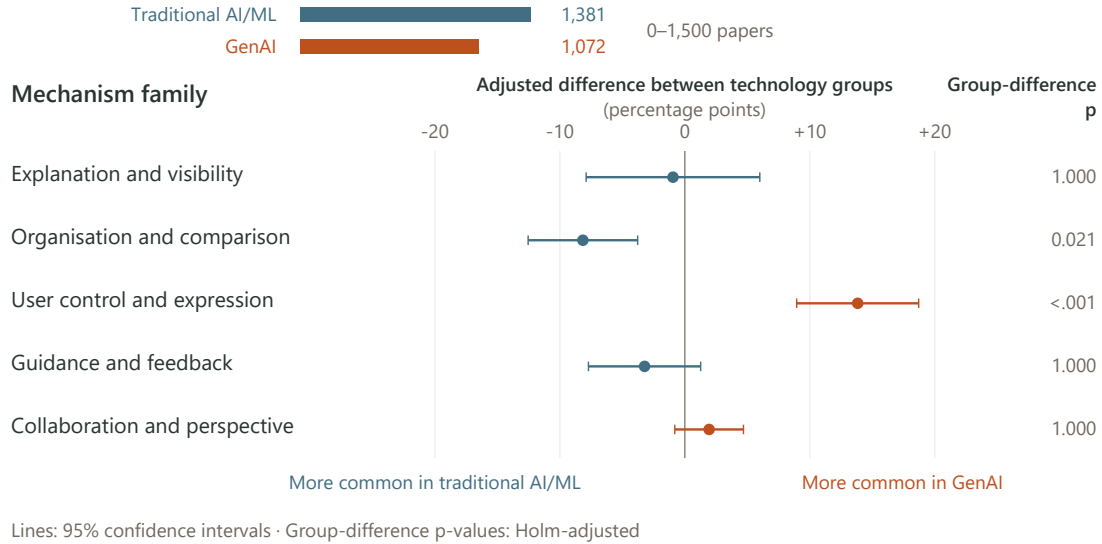


Fig. 10. Between-technology differences in mechanism families among 2,453 papers with usable family coding and classified technology, 2019–2025. Dots show GenAI minus traditional AI/ML prevalence, adjusted for year and venue; lines show pointwise 95% confidence intervals. The p -values test these group differences, with Holm correction across 51 contrasts, rather than change over time. User control retains support in the 2023–2025 and partial-2026 sensitivity windows. Of 3,424 family-coded papers, 544 lack classified technology and 427 classified papers fall outside 2019–2025.

GenAI and 11.2% of traditional AI/ML papers; -6.5; 95% CI -8.3 to -4.7), and no other facet in the family differs by more than 2.9 points. Appendix D gives the annual family profiles (Figure 57).

Collaboration and perspective became more common over time. Holding the technology mix fixed at the original analysis's reference composition (43.75% GenAI), its average annual change was +1.13 percentage points (95% CI [0.53, 1.73], Holm $p = 0.005$). The 2023–2025 sensitivity also showed an increase (+2.84 points per year, CI [1.27, 4.41], Holm $p = 0.007$). Guidance and feedback increased in the primary window (+1.61 points per year, CI [0.72, 2.50], Holm $p = 0.007$), but its recent interval included zero ([-2.69, 6.54]). None of the five tests of change in the GenAI–traditional AI/ML difference survived Holm correction in either window. These temporal estimates (Figure 11) answer a different question from the overall technology-group differences in Figure 10; Appendix D.3 reports both temporal estimands for every family.

Combining mechanisms in iterative work.

Mechanisms can also be grouped by the activity they support: *specify intent*, *represent possibilities*, *evaluate alternatives*, and *act and revise* (Figure 12). Appendix C.1 maps all 110 facet codes to these activities; a code can serve several. Annotations, histories, shared views, and handoffs connect the activities across time and people. The functions describe dependencies, not a fixed order: comparing needs alternatives and criteria, and a comparison may prompt a change to either.

Collaboration and perspective increase across years

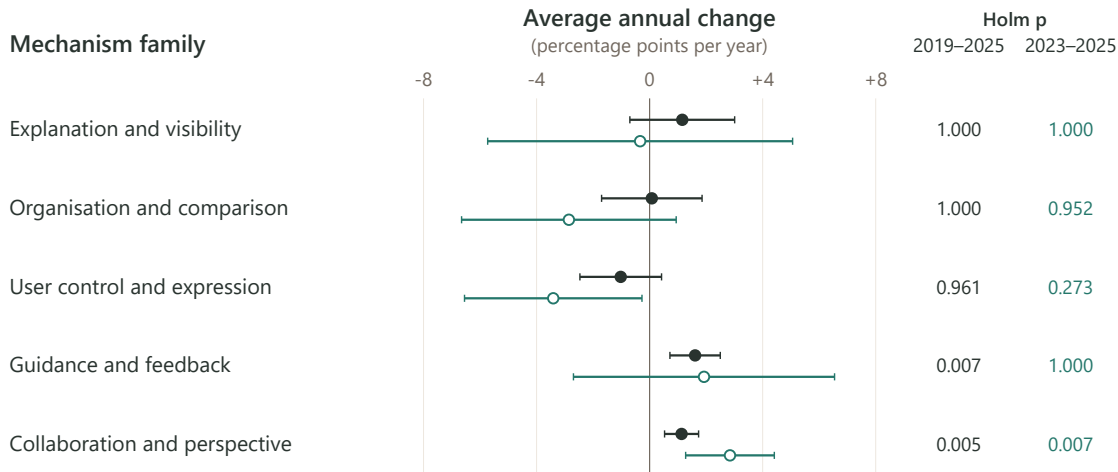
Annual change in coded prevalence, with a fixed mix of AI technologies

Adjusted for venue · 39 venues · Partial 2026 excluded

● 2019–2025 · 2,453 papers

○ 2023–2025 · 1,391 papers

Mechanism family



Lines: 95% confidence intervals · Holm correction across ten temporal tests per window

Fig. 11. Average annual change in coded mechanism-family prevalence: 2019–2025 (2,453 papers) and the 2023–2025 sensitivity (1,391). Dots show percentage points per year; lines show pointwise 95% venue-clustered confidence intervals. The technology mix is held fixed at 43.75% GenAI in both windows. Positive values indicate increasing prevalence. These are temporal slopes, not the overall technology-group differences in Figure 10. Appendix D.3 gives the methods, changing-gap estimates and all ten tests used for Holm correction within each window.

Different mechanisms can serve the same function. Intent elicitation, constraints, and shared criteria frame a task. Structured alternatives and source links make possibilities inspectable; comparisons, explanations, and reflection support judgment; selection, editing, and changed constraints enable action. One mechanism can serve two functions: a comparison exposes options and supports choice, and a constraint can frame a task or redirect it. Guidance can enter at any point, and histories keep context across transitions.

Luminate’s dimensions organize alternatives for exploration and evaluation, with return paths to earlier prompt-specific spaces [129]. AI Chains instead feeds one step’s output into the next step’s input, so people can inspect, edit, and re-run intermediate steps [146]. The order in which a system computes need not dictate the order in which people inspect or repair its work.

An ordered workflow can still require return paths. CollabCoder opens shared comparison after independent coding, then supports discussion, decisions, and grouping while letting people restore their original codes [44]. Crystalline instead builds a route back into the material: a reminder marks related evidence on pages the developer has visited but not yet attended to, and three of twelve participants reported using it to guide their research [80]. CollabCoder sets an order for the work; Crystalline returns people to an earlier step when their comparison may have a gap. Transitions therefore deserve the same design care as individual mechanisms: keep the criterion that shaped an

From task intent to action

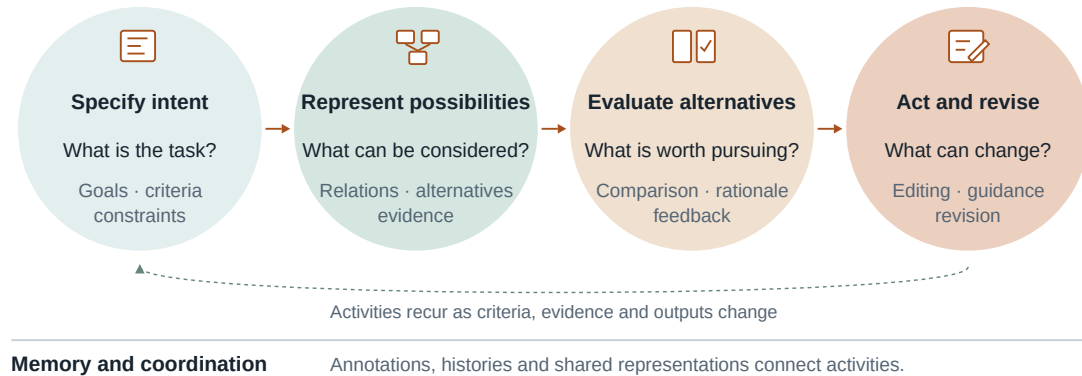


Fig. 12. **From task intent to action.** Four overlapping functions offer an activity-oriented view of the mechanism taxonomy. Memory and coordination connect them across time and people. Arrows show possible dependencies and return paths, not measured transition frequencies. The accompanying source examples illustrate these relationships [44, 80, 129, 146].

alternative, the evidence behind a draft, and a route back to an earlier decision, and test whether people understand a transition, change it, and recover its context.

Scholastic shows why the effect of a control also needs to be understandable. Researchers' codes constrain keyword clusters, while linked document maps, readers, and code views support their interpretation. In a formative study with two researchers, the participant who saw a model update expected documents with the same code to cluster together. They instead remained spread across clusters: the constraint bound keywords, not document membership [50]. The mismatch concerns how the control's effect was understood: it constrained keywords while the researcher expected it to reorganize documents. Exposing a model update alone does not make that relationship clear.

6.3 Revision, preservation, and control

An editable step does not guarantee that people can correct it and keep the later work they still want. Across the cases, the design question is what a human action authorizes the system to reuse or treat as accepted. Table 1 distinguishes saving an old version, preserving valid edits during regeneration, and confirming a result.

Keeping an old version does not carry its edits forward. Kazemitabaar et al.'s data-analysis systems saved earlier versions in branches, but regeneration ignored later manual edits [62]. Participants described avoiding revisions and expecting to redo or reread work; one wanted invalidated parts highlighted instead of regenerated, and another valued the interconnected updates. Recovering an old version differs from preserving the still-valid parts of that work in a new one.

LineageD's pilot showed the cost of predicting a whole lineage at once: correcting one pairing invalidated the rest of the path. The redesign predicts one level at a time, but moving to the next level implicitly confirms pairings the person has not individually confirmed [49]. Plume keeps locked or manually edited text and uses it as context for later generation [79], so a lock preserves wording without limiting its influence on other text or vouching for its correctness. Relating a local step to the whole process is hard in itself: in AI Chains, four of twenty participants asked how a local change would affect the final result [146].

Table 1. **Revision and downstream effects.** What three interfaces preserve, update, or treat as confirmed.

Design choice	Comparison and finding	Conditions and contrary evidence	Design implication
What survives regeneration?	In task decomposition, editing an assumption, plan or code component can regenerate later work while omitting manual edits [62].	The earlier version remains recoverable in a branch; preservation within the revised result is a different property.	Make the fate of earlier edits inspectable when an upstream decision changes.
What does locking text preserve?	Plume retains locked or manually edited dashboard text while using it to inform further generation [79].	An unchanged passage can still influence other text; locking does not isolate its downstream effects.	Distinguish preserving wording from restricting its influence on new content.
What counts as confirmation?	LineageD permits corrections before predicting the next level; advancing implicitly confirms remaining pairings [49].	Explicit and implicit confirmations share the green marking, so the display does not distinguish how a pairing was accepted.	Preserve whether an item was checked explicitly when that distinction matters to later work.

Interfaces promise to keep different things. Automatic propagation can itself help. CoPrompt participants valued automatic coordination, and eight used history and difference views to resolve conflicts or restore versions [38]. Pail’s code differences, summaries, and design panel supported inspection, but participants reported difficulty following cascades across them, less so when they paused to read the changes [148]. Selecting part of an artifact does not by itself fix what stays unchanged: FusAIn leaves unselected image regions untouched, whereas Textoshop can change text beyond the selected passage [91, 109]. Blace keeps earlier states for rollback, and DesignManager records the process behind an asset [83, 147]; both support recovery and inspection of past work, while consistency with a revised goal still requires checking the downstream artifacts. A design should name the property it protects and the change that could violate it, and an evaluation should test whether people anticipate that change, detect a violation, and recover the work they want to keep.

Edits can change the specification, the model, or the output. A correction can change what the person asked for, how the system understood it, or what it produced. FusAIn lets designers compose and revise a visual specification [109], and Falx synthesizes candidate charts from examples, which participants described checking against input values they knew [136]. CoAutoML lets novices inspect and edit the task, inputs, target, and evaluation assumptions it proposes before they confirm a machine-learning task [128]. Plume’s locks retain part of what the system produced [79]. Outside generative chat interfaces, corrections can also update the model: SmartEye uses a corrected crop as a positive training example for later recommendations [87], and DeepSI fine-tunes BERT from spatial feedback and recomputes its document map, though its authors note limited visibility into the model’s internal state [13]. The design question is what a correction changes and which later operation uses it, not whether every control comes before generation.

Generating alternatives and their criteria.

When a system generates the criteria as well as the alternatives, people must judge both. In Luminate, selecting an axis rearranges the responses, whereas adding a dimension revises the responses themselves [129], so the writer in our opening example judges the dimensions as well as the ideas. VIVID generates expected answers and rubrics along with candidate dialogues, and instructors inspect and edit them [29]. Design-rationale research already links questions, options, and criteria [89], and conventional tools sort alternatives into categories that designers choose. Generated

Table 2. Selected AI-involved comparisons of explanation and practice designs.

Comparison	Assessed result	Conditions
Predicted-misconception explanations [21]	Better immediate exercise-prescription decisions after AI removal than unilateral explanations or no AI	Advantage over random foils unresolved; no delayed test
Required reflection after AI feedback [7]	2.0 versus 3.4 problems completed in eight minutes; transfer and code-evaluation advantages unestablished	Fixed time budget; fewer attempts do not establish learning harm
QuizBot versus flashcards [116]	Higher recognition (87.2% versus 65.8%), with an advantage at two months	Equal repetitions; 46 versus 14.3 minutes of practice; dialogue bundled with other support
Adaptive versus generic video recommendations [73]	Small algebra-ability advantage during normal schooling	Low optional uptake; policy effect; closure-period advantage unresolved

criteria raise a narrower question: can people revise a criterion while holding the alternatives fixed, or does the revision also replace what they are judging?

Generated criteria can improve comparison. Selenite gives readers generated overviews of criteria and labels the relevant passages; in a twelve-person paired comparison with unaided reading, participants produced more valid criteria with the overview (12.93 versus 6.79), with higher precision and recall [81]. A criterion can also be easy to act on and still miss the quality it stands for. RECAST lets people revise text against a toxicity model; revisions lowered the model's toxicity scores without reliably improving human ratings of toxicity [144]. When both the model's response and human judgment matter, the evaluation has to measure both.

A label can also leave the question itself ambiguous. In a prior sensemaking study (Stream 2), an unexplained "trust" lens led some researchers to consider a person's trustworthiness and another to consider that person's trust in sensor data [35]. The same displayed criterion can therefore invite different judgments; understanding its use requires following what people take it to mean.

6.4 Comparing explanation and practice designs

Explanation content, required reflection, dialogue, and optional recommendations change different parts of practice. Table 2 links four AI-involved comparisons to their assessed outcomes.

The practice cost of reflection does not explain every null learning result. In Xiaomai's advanced-mathematics study, requiring error reflection produced neither a detected learning advantage nor a detected difference in attempts [37]. Together with the fixed-time comparison above, this makes the content and uptake of reflection worth examining alongside the practice it displaces.

Explanation design also concerns sequence. Incremental XAI supplies factors progressively during model-prediction practice, then tests prediction on unseen cases without the explanation display [16]. This separates the presentation of an explanation from what a person can do once it is removed.

Prior learning research separates creating an explanation from using it. In a randomized study of 120 learners, everyone wrote subgoal labels; those who then practiced with their own labels scored 21.6 of 25 on an immediate test, against 17.1 with expert labels and 18.8 with none. Instructional materials were removed for the test, while the programming environment remained available [90]. This comparison sharpens ExPeerience's question (Section 5.2): which part of the task should the person learn to perform? Learning research has long stressed what support asks learners to do [28, 113]; the interface choice is what to generate and what to leave for the learner.

CoWriting Kazakh tests a related division of work: children either try writing a word in the new Latin script before seeing the robot’s correct spelling, or write it in Cyrillic and let the robot convert it. Both conditions show the robot’s model spelling. Across 67 children, letter knowledge increased during the single session, but the study found no overall advantage for either conversion condition [118]. The design question is which operation the learner needs to practice and which feedback makes that practice useful.

NoteCopilot makes the amount of supplied structure a concrete comparison. In its thirty-person, within-person study, intermediate assistance yielded higher immediate understanding scores than either minimal or automated assistance when learners first answered without their notes [27]. The packages changed both the supplied material and the work of organizing it. This supports examining the division of work, without identifying engagement as the cause or establishing a delayed benefit.

Writing feedback can describe a revision or judge its quality. ArgRewrite classifies the purposes of a writer’s earlier revisions without judging their quality, and asks writers to agree or disagree with those labels before and after revising again [1]. ArgueTutor pairs claim and premise labels with recommendations based on their balance [135], and LegalWriter adds advice on argument components while leaving the semantic fit between claim and conclusion to the writer [140]; both improved the quality of text written with them over their comparison interfaces. A description of a revision and advice about quality make different assertions, and one interface can offer both. A learning claim needs a test of the judgment or revision skill the feedback is meant to build.

6.5 How prior knowledge informs design

Prior knowledge enters a design through the material it supplies, the activity it asks of a person, and the measures used to assess it. The lineage comparison asks where those connections are made explicit and where a further inference is needed. The following cases trace a rationale into an interface feature, distinguish an activity indicator from a learning assessment, and examine whether evidence from an earlier task supports the new design’s claim.

The source-use map (Figure 14) links 1,697 matched lineage cards to 43,733 source–paper pairs in the current analysis. A source recorded in two papers contributes two pairs. The records distinguish concrete uses in a design or measure, conceptual influences, and citation-only mentions. These roles describe how prior work enters a paper; they are not a ranking of theoretical quality. Papers without matched lineage evidence remain outside this comparison.

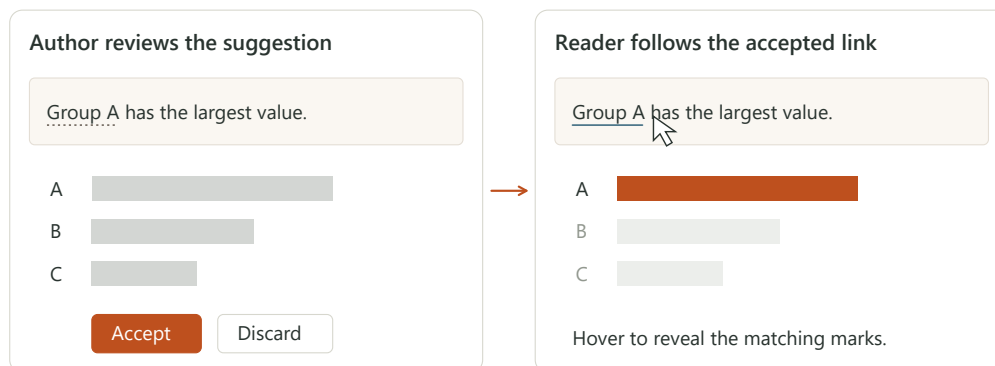
A rationale can leave the design choice open. DataDive draws on statistical thinking and critical reading to motivate contextualizing numerical claims. Its three design goals come from a seven-person formative study; the candidate contexts and ranking criteria are author-specified choices [65]. This establishes why context matters while leaving which generated questions help readers reason as an empirical design question. Its 21-person comparison found no detected advantage in data or data-interpretation recall over ordinary search; readers recalled more facts from outside the text in the baseline condition. The rationale, the interaction, and the assessed consequence support different parts of the account.

Making correspondence inspectable. Kori turns the concern about split attention into a visible link between prose and chart marks [74]. The system proposes links, and authors preview, accept, discard, or draw them by hand; once a link is accepted, hovering over its phrase highlights the chart marks it refers to (Figure 13). The author checks each correspondence, and the reader uses it to interpret the data.

Operationalizing a theory is not testing learning. QuizBot cites ICAP [28] as its learning rationale, whereas the Khanmigo methodology turns ICAP into an LLM-judged indicator of cognitive engagement [116, 132]. Deeper use of the theory does not bring stronger evidence of learning: QuizBot tests recognition two months later, while Khanmigo

Kori makes text–chart links inspectable

AI proposes a connection. The author decides whether to keep it.



The author checks the correspondence; the reader uses it to navigate the data.

Fig. 13. **From an AI suggestion to an interactive reference in Kori.** The author reviews a proposed text–chart connection. After acceptance, hovering over the linked phrase highlights the matching marks by fading the others [74]. Schematic redrawn from the paper’s interface description, with illustrative data.

tracks correctness on the next item a student attempts alone. A direct test in diagnostic reasoning makes the distinction sharper: problematizing elicited more constructive responses than structuring, yet no condition advantage on diagnostic strategies was detected in subsequent cases after the pedagogical module was removed [46]. Both groups received active scaffolds, and the tests occurred in the same session. A theory can identify a promising activity; observing more of it still leaves its contribution to capability to be established.

Prior knowledge can shape the evaluation as well as the design. Sibyl adopts Doshi-Velez and Kim’s application-grounded evaluation [34, 156], testing explanations with screeners in child-welfare work. Screeners often rated case-specific explanations helpful and found other forms less useful during those cases, and their feedback exposed problems of terminology and presentation.

A theory’s evidence travels with its comparison. Codigen cites cognitive mapping to motivate active learning and supplies generated programming material on an editable whiteboard [36]. The cited medical study compares maps with notes organized around the same five problem-solving elements, with the assigned support available during the diagnostic case [138]. It does not compare self-authored with generated material. Codigen’s novice pre/post quiz gains likewise do not isolate the whiteboard’s contribution. The connection suggests a design, while testing its learning rationale would require comparing how learners construct and use the material.

These cases locate different gaps in the connection from theory to evidence. Kori makes a design response to split attention inspectable; the ICAP comparison separates a theory-motivated design, an engagement indicator, and evidence of later capability; Codigen illustrates a new design inference that the cited comparison did not test. The useful question is where the explanation needs more evidence, rather than how many theoretical sources a system cites.

Supplied material can support learning through its use. Learners need not build the material they learn with. VocabEncounter supplies phrases in context and reports better vocabulary two days later than its word-only condition

Table 3. Measurements of process, artifact, and learning. Timing and available assistance are specified separately for each measure.

Outcome	Assessed in the study	Reported by participants
Process	Task time, action sequences, revisions, errors	Perceived effort, workload, usability, agency
Artifact	Correctness, completeness, external quality ratings	Perceived quality, usefulness, originality
Learning	Scored understanding, skill, retention, transfer	Perceived learning, confidence in ability, self-efficacy

[5]. Rehearsal supplies conversational alternatives, practice, and feedback; its participants used more cooperative strategies in a later unaided conversation than participants given the same static materials without interactive practice [121]. In a study of AI assistance for incidental learning, explanations without a recommended answer improved immediate concept learning over minimal feedback [42]. Supplied material also leaves room for different uses: CoNotate participants reported both running its suggested queries and borrowing their vocabulary to write new ones [104]. The design question is which operation the supplied material helps people perform, and whether the study tests that ability.

6.6 What capability the evaluation establishes

Three questions help interpret the comparisons: what did people do, how good was their work, and what could they do on a subsequent attempt? The process, artifact, and learning (PAL) framework (Table 3) records those targets alongside the comparison, timing, assistance, and method of assessment. It distinguishes observed or externally assessed outcomes from self-report. Rating a finished artifact later still measures that artifact, rather than what its author has learned.

Product and capability measures answer different questions. In BELVEDERE, the maps showed more explicit links between evidence and hypotheses, but no difference in final prose conclusions was detected between conditions, and students could still consult their maps while writing [130]. Both outcomes describe products made with support; neither tests what students can do without the maps.

A benefit depends on the comparison and the situation. Cognitive forcing asks people to engage more deliberately with AI advice. Among 199 participants, fully correct decisions averaged 35% with simple explanations and 33% with forcing, a nonsignificant difference; on items with incorrect AI advice, forcing raised correctness from 3% to 9%, while the no-AI condition reached 18% [20]. Adaptive help raises the same issue: predicting whether a person will answer correctly differs from showing that advice based on that prediction helps them [85, 127]. Appendix A.7 gives the learning and duration profiles of the reviewed studies.

Assistance also defines the capability being tested: independence from a helper can coexist with continued use of a device. We return to that design choice in Section 7.3.

6.7 Performance and subsequent capability

An interface can improve one assessed outcome while leaving another unresolved. Table 4 distinguishes comprehension, revision quality, enacted behavior, and application of supported skills.

These comparisons apply Norman’s distinction between performance with an artifact and the capability attributed to the person [100]. DBox’s decomposition and feedback also improved performance on an immediate similar problem solved without assistance, although effort rose [86].

Assessment conditions specify the capability demonstrated. Among the 396 AI-involved sources in the outcome appraisal, 223 reported a separate capability assessment, 31 assessed capability in a later session, 71 used new material or a changed task, and 67 reduced or removed a named training aid. These overlapping features describe the selected appraisal set (Appendix A.8).

Table 4. Assessed outcomes in four AI-involved comparisons.

Comparison	Assessed result	Conditions
CodeVoyager versus same-model chat [66]	Higher immediate closed-book comprehension (8.06 versus 5.44/20); no detected assisted-completion or accuracy advantage	16 developers; interface package; no delayed test
FeedbackWriter versus usual TA feedback [82]	Better revisions ($d = 0.5$); knowledge-test advantage unestablished	Randomized 354-student trial; optional posttest; test-time assistance unclear
Rehearsal added to static training [121]	Better immediate conflict-response behavior; recall and recognition advantages unestablished	Behavior and recall assess different capabilities
CodeTailor versus generated solutions [53]	Better immediate unsupported application of supported elements; total posttest advantage unestablished	Selected 14 of 18 participants; tested elements depend on practice difficulty

An assessment's task, timing, and available support each change what it shows: an immediate new problem can test application without testing retention, and a later test can still allow help. WriteAid tested pupils' writing two to three weeks later without AI or teacher support, but on the same topic and without an untreated comparison or a pretreatment skill measure [99].

The promised use determines the informative test. Immediate performance, later retention, and application to new material need not move together. Two classroom experiments with structured LLM reflection found no significant advantage on exams two weeks later [71]. Preference is not capability either: across three AI Margin Notes studies of reader interfaces, note selection, and human and AI note-taking, comprehension two hours after reading did not differ significantly, even where preferences or task demands did [60].

An informative evaluation therefore starts from the design's intended use. An explanation meant to prepare a new decision needs a decision task; feedback meant to change the next attempt needs that attempt observed; a practice partner meant to build interpersonal skill needs evidence of enacted behavior, while recall answers a different question. Effectiveness reviews already show that scaffolding can improve cognitive outcomes [11]; the comparisons here show which alternative and which outcome a designer can use as evidence.

7 DISCUSSION

Semantic scaffolding changes more than the amount of help a tool provides. It gives people material to think with, a way to judge that material, and particular work to do themselves. We draw three design positions from the findings: people need to be able to question the structure a tool supplies; evaluations need to distinguish better work from learning; and people should be able to reshape support as their work and capabilities change. The taxonomy makes the available choices visible. The comparisons explain why their value depends on the judgment or practice they make possible at a particular moment.

7.1 People need to judge the scaffold as well as its output

An AI tool can propose both an answer and the grounds for judging it. Luminate generates dimensions for comparing ideas; VIVID generates rubrics and expected answers alongside candidate dialogues [29, 129]. This can make a difficult comparison more tractable: Selenite helped readers identify more valid criteria [81]. But a usable criterion can still be a poor account of the quality people care about. In RECAST, reducing the model's toxicity score did not reliably improve human ratings [144]. The sensemaking study adds a semantic problem: the same "trust" label referred to different

relationships for different participants [35]. People need to decide both what a criterion means and whether it captures the quality they care about.

Design rationale connects questions, options, and criteria [89]. Generative interfaces bring a new version of this problem into ordinary use: the system can compose the options and the criteria at the same time. People need to distinguish “Which option meets this criterion?” from “Is this the right criterion?” The interface can support that distinction by showing where a criterion came from, letting people compare different standards, and allowing them to revise a standard without automatically replacing the alternatives being judged.

Editing controls matter insofar as they make these choices practicable. Plume preserves selected wording during regeneration; the reviewed data-analysis systems can retain an earlier version while omitting its manual edits from regenerated work [62, 79]. A person reconsidering an assumption needs to see what that changes and keep work that still holds. Supporting that judgment means preserving valid contributions and making the consequences of a changed assumption inspectable.

7.2 Evaluate the work, its results, and what people learn

Three questions organize evaluation: what did people do, how good was the result, and what did they learn? The process, artifact, and learning (PAL) framework in Table 3 keeps those questions distinct. Process evidence records actions such as checking, explaining, and revising, together with their time and effort. Artifact evidence concerns the resulting draft, program, or decision. Learning evidence concerns a person’s capability on a subsequent attempt. Norman’s distinction between the person–artifact system and the person’s activity explains why success on one cannot answer the others [100].

The theory lineage helps explain why a feature should change a particular activity and where that explanation still needs evidence. A system can contribute by making a theoretically important activity possible or changing how people and AI share it, even with familiar controls. For example, supplied and learner-constructed criteria can organize the same alternatives while assigning different judgment to the learner. The design claim specifies what should change; PAL keeps the evidence about activity, resulting work, and subsequent capability distinct. This makes it possible to ask whether a study supports the proposed explanation or leaves a connection unresolved.

The distinction also concerns what an assessment means. In Inie et al.’s 80-paper HCI scoping sample, studies often leave critical thinking undefined or assess it through self-report [58]. Feeling more reflective, producing a stronger argument, and becoming better at evaluating an argument are separate claims. A learning-oriented tool needs to name the capability it is designed to develop and show how its assessment tests that capability.

The reviewed comparisons make the distinction consequential. CodeVoyager improved immediate closed-book comprehension without a detected advantage in assisted completion or accuracy; FeedbackWriter improved revisions without an established advantage on its optional knowledge test [66, 82]. Rehearsal improved enacted conflict-response behavior while recall and recognition advantages remained unestablished [121]. These are different benefits, and the appropriate learning measure depends on the capability the tool is meant to develop.

Observing the process helps explain the benefit and its cost. Required reflection displaced some practice within a fixed interval; QuizBot’s equal-repetition comparison allowed substantially different practice times [7, 116]. In the algebra-video trial, optional uptake was low: the tested intervention was the offer of adaptive recommendations, not assigned video watching [73]. Evaluating all three questions means following the work people actually perform, judging its result, and testing the intended capability. It does not require one combined score or assume that more interaction produces more learning.

7.3 Adapt support as people develop

Learning need not culminate in working without tools. Pea distinguishes scaffolding intended to fade from continuing distributed intelligence [108]; Krakauer asks what capability remains when an artifact's immediate advantage is removed [70]. Both direct attention to what the person gains. In vocational prompting, device-supported task correctness counted differently from asking a person for help [25]. The intended independence concerned human assistance, while the device remained useful.

This permits a deliberate division between useful automation and meaningful practice. Reiser distinguishes structuring a task from making its problems available for reasoning [113]. ExPeerience generates problem contexts and explanations while leaving SQL construction to the learner [153]. Rehearsal supplies a counterpart and feedback while the person practices responding [121]. Generating material can create an occasion for human activity as well as replace one. Designers should choose which capability deserves practice, then arrange the surrounding support to make that practice possible. The test is whether people become better at the chosen activity, with the resources their future work will actually provide.

Support should therefore respond to two kinds of change: the decision a person faces now and the capability they are developing across attempts. Expertise-reversal research gives a reason to distinguish them: guidance useful to a novice can become redundant as knowledge grows [61]. Malleable interfaces provide ways to reshape the representation, while persistent organizing rules can retain what the person has already established [63, 93, 95]. Maru also shows how a rule from an earlier subtask can persist where it no longer fits [63]. Keeping a useful structure therefore needs to include ways to revise or retire it. For example, a tool could initially supply criteria, later invite the person to revise or propose them, and retain access to examples throughout. This is a design proposition to test over repeated use. Its purpose is to change which judgments people can make, while keeping the support those judgments still require.

8 RESEARCH OPPORTUNITIES

The review opens four questions for research: which parts of a scaffold people should construct themselves, which activities account for learning, how judgments develop across people in real work, and how everyday tools can make room for learning. Each calls for studying a particular design choice rather than treating semantic scaffolding as a single intervention.

8.1 Decide which parts of a scaffold people should construct

AI can supply a comparison, a plan, or a rubric almost immediately. The design question is how much of that structure to provide and what to leave people to construct. Supplying criteria can help people notice distinctions they would otherwise miss, as in Selenite [81]. Constructing or revising criteria may instead engage people in deciding what matters. This is a concrete instance of the assistance dilemma: support can make progress possible while also taking over an activity the person needs to practice [69, 113].

Compare the same alternatives under three designs: supplied criteria, criteria people can revise, and criteria they construct with access to examples or feedback. Hold the alternatives fixed so that revising the standard does not also regenerate the objects being judged, as can occur when adding a dimension in Luminare [129]. Vary prior expertise and the stage of work. Examine which relevant distinctions people notice, how they justify a choice, and whether they can build a useful comparison on a new case. The aim is to learn when ready-made structure enables judgment and when participating in its construction is itself the valuable activity.

8.2 Identify which activities help people learn

An interface often bundles an explanation, practice, feedback, and extra time with the material. The theory lineage can help identify which activity the design expects to matter and what observation would challenge that account. When a study finds a benefit, which part made the difference? QuizBot's comparison held repetitions constant while practice time differed substantially; required reflection consumed some of a fixed practice interval [7, 116]. These comparisons motivate studies of what people do with support, beyond whether one interface package outperforms another.

The diagnostic-reasoning comparison shows why this matters: more constructive responses did not establish a subsequent strategy advantage [46]. This is useful for theory-informed system development: an activity can be elicited while its contribution to capability remains unresolved under the tested conditions.

Separate a focal activity from the rest of the interface: for example, compare seeing an explanation with predicting an answer before seeing that same explanation. Keep the task content, available help, and feedback comparable. Test equal-time and equal-repetition conditions where the activity changes the amount of practice. Observe uptake and reasoning, assess the work produced, and test the intended skill on new material immediately and after a delay. For optional support, distinguish the effect of offering the feature from the self-selected behavior of using it [73]. This would test whether adding the activity improves the intended skill under each practice budget, and whether gains persist on new cases with the resources available in ordinary use.

8.3 Study how people form and share judgments in use

Interactive systems offer a way to study how reasoning develops around a representation. Luminate makes proposed criteria available for revision; CollabCoder brings individual interpretations into shared discussion [44, 129]. Distributed cognition provides a basis for following the work across people and representations [48]. Observe when people adopt a criterion, contest it, or carry it into a later decision, and examine how deadlines and the demands of producing an artifact shape those choices. Interface traces need observation and participants' explanations to reveal why a judgment changed.

Handoffs make this question especially concrete. Compare a coaching summary alone with the same summary carrying the learner's priorities, evidence, and open questions, allowing equal preparation time [17, 55]. Follow what the coach infers, what the learner corrects, and which issues shape the next meeting. Analyze the coach-learner group together; assess learners directly when the claim concerns their learning. The opportunity is to understand which representations help people maintain and repair a shared interpretation as work moves between them.

8.4 Create learning opportunities within everyday tools

Tools for getting work done could also offer people opportunities to develop capabilities they want to retain. An explanation before a decision, a comparison between alternatives, or feedback on a proposed revision can make a moment of ordinary work available for practice. Incidental-learning research found an immediate concept-learning benefit from explanations without a recommended answer, compared with minimal feedback [42]; ExPeerience illustrates how generated material can leave the target activity with the learner [153]. These cases motivate embedding opportunities to reason within work, instead of treating learning only as a separate training phase.

Develop this approach around a capability chosen with the user: recognizing a misleading comparison, judging a source, or repairing a program. Offer a brief prediction, explanation, or correction at a relevant decision, with control

over when to participate. Malleable interfaces and persistent information architectures offer ways to change that invitation without discarding the person’s working representation [63, 95].

Compare fixed support, user-directed changes, and changes the system proposes from observed difficulties over repeated use. Let people accept, defer, or reverse a proposed change: declining practice during an urgent task need not mean that they have mastered the skill. A useful moment might be a disputed criterion, an unsupported claim, or a recurring error, when the value of examining it is apparent. Assess interruptions, work quality, uptake, and performance on new cases immediately and after a delay. Remove the learning prompt while retaining resources that normally belong to the task. This would test when adaptation creates useful practice, when it becomes an interruption, and which support should remain as capability grows.

9 LIMITATIONS

Coverage. We searched 40 venues over 8 years, which favors published, English-language research [6, 76]. Technology comparisons leave out 645 papers whose AI technology could not be determined, and each analysis uses the papers coded for its fields (Table 9). The theories and empirical studies we compare against are a selection, not a census of every relevant tradition.

Reading and coding. Language models read each paper against a fixed codebook, one model per paper, and grouped the coded descriptions into categories. People have not yet checked the coding or the cross-case interpretations. Models can miss evidence shown only in figures and can share biases, and other analysts could draw the boundaries between closely related mechanisms differently.

Effects. Studies differ in tasks, participants, comparators, outcomes, and duration, and many compare whole systems. Most coded changes in how people work rest on the authors’ own attribution rather than a measured contrast (75.8% of coded process shifts). The GenAI and traditional AI/ML comparisons are associations among published designs, not effects of the technology. The design space cannot rank mechanisms by effectiveness or estimate an overall learning benefit.

10 CONCLUSION

AI tools increasingly supply the structures people think with: criteria, examples, plans, and records of earlier choices. Our design space maps these semantic scaffolds across domains and connects each to the work it organizes and to what studies found. Generative-AI papers more often concern producing artifacts and include user-control mechanisms. What people can change matters, whether a specification, the AI’s interpretation, the criteria, or the result, and support can reach a learner through peers, teachers, and coaches. Supplied material can support learning through the way people use it, and added interaction does not by itself make people practice the intended skill. The design space offers concrete alternatives for helping people develop capability within everyday work. The next challenge is to understand when to change that support and what should remain as people learn.

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A EVIDENCE EXTRACTION AND CODING

This appendix holds what a reader needs to check or reuse the coding: the fields of each evidence card, how the closed and open fields were coded, the agreement checks, the data behind each analysis, how sources for the outcome comparisons were chosen, and how cited sources were matched.

A.1 Card Fields

Stream 1 mechanism cards (Table 5) describe the design, and theory-lineage cards (Table 6) describe its use of prior work. Each Stream 2 source has a knowledge-claim card (Table 7). Table 9 distinguishes the available evidence for each analysis. The sectioned layout follows Model Cards [96], and AI-assisted design-card generation is a precedent for turning source findings into cards [125].

Table 5. Stream 1 mechanism card: work and human activity first, then scaffold design and evaluation.

Card section	Fields recorded
Workflow changes (RQ1)	Process before and after support; mechanism credited; attribution basis; source of the baseline.
Human activity (RQ1)	Activity externalized to the tool or representation; activity retained by people; work still required.
Scaffold design (RQ2)	Author term; facet and value, with their merged codes; what makes each mechanism separate; supporting passage and page.
Learning and support (RQ2)	Learning measurement, direction, and evidence; reliance; changes in support and their grounds; observation horizon.
Measures and comparisons (RQ2)	Recorded outcome layer (process, artifact, learning); author claim, claim scope, contribution type, study design, participants, comparison, statistics, and claim-evidence alignment.
Conditions and limitations	Author-stated limitations and boundary conditions.
Recorded codes	Work domain, intent, expertise, capability gap, support properties, AI involvement, and technology era.
Source	Paper ID, DOI, venue, year, page count, the PDF read, the full-text decision, a map of relevant pages, and codebook notes.

Table 6. Stream 1 theory-lineage card: how the paper uses the sources it cites (RQ2).

Card section	Fields recorded
0. Identity and position	Link to the mechanism card; overall position toward prior work; position summary; human-facing rationale; lineage depth.
1. Documented sources (RQ2)	One entry per cited work: citation identity, relation to the focal paper, use, enactment, and supporting passage and page.
2. What was read	Introduction and related-work pages; design or evaluation pages checked for application of a cited idea.
3. Synthesis note	A short account of what the paper's use of prior work shows.

A.2 Closed Fields

Closed fields take values from a fixed list. Intent records whether the support targets immediate performance or sensemaking, learning or skill development, or both. Measurement records what the evaluation assessed, so an intention

Table 7. Stream 2 card: one to eight field-level claims, linked to the research questions they inform.

Card section	Fields recorded
0. Identity and position	Card and work IDs; publication details; discovery routes; contribution types; position summary; overall uncertainty.
1. Page-grounded claims	One entry per claim: author wording and reviewer interpretation; question links; applicability; affected human work; outcome evidence; evidence basis; counterevidence and limits; supporting excerpts; uncertainty.
2. Evidence store	Verbatim excerpts with pages and locators; links recording which claim each excerpt supports, its role, and the reviewer paraphrase.

to support learning stays apart from a measure of it; the direction of a result and its observation horizon are recorded with the measure. Support properties are coded separately and can co-occur: structuring, problematizing, contingency, adaptation, fading, responsibility transfer, and distributed support. Contingency responds to the person's activity or need. Fading reduces support as competence grows; a fixed reduction on a schedule is not by itself fading. Responsibility transfer concerns who performs or regulates the activity, not whether the interface remains available. Technology is classified once per paper, from the AI in its system, an experimental condition, training, or evaluation, so a paper classified as generative AI can still contain mechanisms that involve no generation.

A.3 Open Fields

The 21 open fields hold 142,877 observations in the papers' own terms. Equivalent descriptions within a field were merged into 1,290 codes, and the codes were grouped into 223 categories and 92 families across 18 fields, with subcategories only where they add a distinction. Each grouping keeps links to its observations, and fields are never pooled. Codes that record missing information, such as an unspecified baseline or an unreported measure, stay apart from substantive categories. When one free-text value lists several limitations, its single code summarizes them rather than listing each. Appendix B defines the families and categories of the six fields the framework uses.

A.4 Agreement Checks

To test the open-code grouping, a separate model pass coded extracted observations with the existing codes hidden, and we compared the two groupings field by field (Table 8). An observation entered the comparison only when its identifier, source card, and field value matched exactly. Two codings need not use the same words, so we compared partitions: whether each pair of observations shares a code in each coding. The adjusted Rand index (ARI) corrects pair agreement for chance, and positive-pair F1 summarizes overlap among the pairs either coding groups together; neither treats one coding as correct.

Raw agreement is high because both codings keep most pairs apart: for facet, 19,083 of 19,503 pairs are apart in both and 32 are together in both. The blind pass used fewer codes than the existing coding in five fields and more in learning evidence. It reported medium or high uncertainty on most assignments in five fields (for example, 659 of 695 values) and on 7 of 24 in learning evidence. The learning-evidence row covers 24 of 48 existing observations; the other 24 share a repeated not-measured value and fell outside the comparison's selection rule.

A separate check asked only whether AI was involved: three fresh model sessions labeled 532 title-and-abstract records added in September and gave the same label to 529 (99.4%; $\kappa = .988$).

Table 8. Blind open-code comparison on exactly matched observations. Codes gives existing / blind code counts; Rand is raw pair agreement, and pair F1 compares the pairs grouped together. Unmatched counts existing observations outside the comparison. Papers overlap across fields, so the 1,571 matched observations are not 1,571 distinct papers.

Field	Obs. (papers)	Unmatched	Codes	Rand	ARI	Pair F1
Externalized to tool	110 (110)	18	64 / 46	96.2%	.167	.186
Facet	198 (48)	15	132 / 76	98.0%	.132	.142
Process before	125 (48)	15	79 / 34	95.8%	.445	.462
Process after	419 (160)	0	262 / 60	97.0%	.079	.088
Value	695 (144)	0	517 / 95	98.1%	.082	.087
Learning evidence	24 (24)	24	5 / 16	50.4%	.108	.217

Table 9. Data behind each analysis. Counts refer to the stated unit.

Data	Count and unit	Analysis
Stream 1 corpus	4,638 papers on 28 September 2026; 475 non-AI comparison papers outside the systematic counts	Corpus counts (Section 4)
Main figures	4,133 papers (4,162 records) with usable coding	Annual, technology, and mechanism figures
Closed fields	2,891 papers coded by 28 August 2026; 301 measure learning	Intent, learning-measurement, and process counts
Open fields	142,877 observations in 21 fields from the 3,295 papers coded by 28 August 2026, of which 2,847 remain in the corpus	Design space and framework values; counts of papers carrying a value
Complete mappings	2,315 papers with complete facet-value codes; 2,311 also with remaining-work codes	Mechanism-by-work matrix (Appendix C)
Theory lineage	1,697 PDF-matched lineage cards; 43,733 recorded source-paper pairs	Roles of prior work (Section 6.5)
Stream 2	512 sources; 2,867 claims	Prior findings and theory
Citation network	4,751 papers from both streams	Citation atlas and co-citation map (Section 6.5)

A.5 Validation Status of the Readings

For part of the evidence, the records of the original reading runs have not been fully validated, and we mark that evidence for provisional use. On 26 September 2026, such evidence appeared in 319 of the 1,076 learning papers and 1,252 of the 4,086 papers behind the main figures. The counts mark incomplete validation records, not errors, and the marked papers stay in the analyses.

A.6 Data Used for Each Analysis

Each analysis uses the papers coded for the fields it needs (Table 9).

All 2,847 open-coded papers have complete domain and retained-work codes; 532 lack complete facet-value codes. The closed-field population includes 44 papers outside the open-field profile and two with unresolved support-property coding.

A.7 Learning Measurement

Among the 2,891 closed-field papers, 301 (10.4%) measured learning. Two-thirds of those (199, 66.1%) measured it within a single session; 35 used repeated sessions, 38 followed people over weeks, and 25 measured learning in deployed use.

A.8 Selection of Sources for Outcome Comparison

The study comparisons in Section 6 draw on sources chosen for their outcome evidence. A learning-scope assessment of 4,714 records proposed 1,226 within scope. We appraised the 422 of these whose cards recorded that learning was measured (twelve more had been appraised earlier and were analyzed separately), plus 14 performance cases found among 53 descriptions that needed interpretation: 436 records, or 434 distinct PDFs. The 190 sources flagged as having a documented capability comparison entered the detailed synthesis.

A.9 Source Matching and Coverage

A.9.1 Source-use coding. Each theory-lineage record links one cited work to its role in one citing paper’s design, argument, or measure (Section 6.5), so a work cited by two papers yields two records.

A.9.2 Citation metadata. The bibliographic map joins the two streams by DOI or Semantic Scholar paper ID and counts a paper found in both once. It uses Semantic Scholar data retrieved on 9 September 2026 for 4,331 Stream 1 papers and 448 Stream 2 works (28 in both). Reference lists for 3,233 papers name 106,523 cited sources; two with conflicting identities are withheld. Of the remaining 106,521, 5,748 are cited by at least five review papers and 157 by at least 40 (Figure 15). A reference list shows that a paper cites a work; its lineage card shows how the paper uses it.

The co-citation map counts each citing paper once per source and keeps sources with at least five citers and pairs with at least three shared citers. It weights a pair by association strength: the pair’s shared citers times the number of papers, divided by the product of the two sources’ citer counts. Louvain clustering (seed 42) forms the neighbourhoods, and a maximum spanning forest supplies the drawn links; stem, branches, and leaves mark how widely a source is shared. Less-cited sources sit near their most shared neighbourhood without joining it. The interactive atlas supports zooming into a neighbourhood and inspecting each source.²

A.9.3 Matching and unavailable sources. Cited works without a DOI are matched by title and year. Title variants can split one work into several identities, which overcounts distinct works and undercounts shared references; descriptive labels and conflicting identities stay out of the shared-reference map. Papers without a reference list, and full texts we could not obtain or read, are outside the source-use analysis.

B THE FRAMEWORK’S CODEBOOK

The framework combines mechanism design with five dimensions of human work and evidence. Tables 10 and 11 define those five dimensions; Appendix C.1 gives the mechanism categories and examples. Counts use the 28 August 2026 coding populations in Appendix A.6; a paper may carry several values. Each group names the paper denominator for its percentages. Daggers identify absent or unspecified values, as defined in each row.

B.1 What follows for the person, the work and the evidence

²Interactive citation atlas: the interactive citation atlas .

Table 10. The three dimensions about the person: what the person still does, keeps, and hands to the tool, with the share and number of papers carrying each value. A dagger marks the absence of a value.

Dimension and value	Definition	Share	Papers
What the person still does (n=2,842 papers)			
The task the tool did not do	Work of the task itself that the system does not do: performing the substantive task, making sense of what is shown, and checking and repairing what comes back.	51.1%	1,451
New work the tool created	Obligations that exist only because the system exists: feeding and instructing it, and operating and coping with it.	29.9%	851
The whole task kept by the person	The person feeds the system and judges what comes back, across three or more stages, with no stage removed.	10.6%	300
Final authority kept by design	The final decision, the reliance decision, values-laden judgement and accountability stay with the person as a matter of design or law, not as a step in a workflow.	8.3%	237
Duty asserted, nothing specified†	The value says responsibility remains with the person without naming any obligation.	0.1%	3
What the person keeps (n=2,847 papers)			
Judgement and decision	Interpreting and inferring, deciding and authorising, setting direction, bringing knowledge.	48.0%	1,367
Doing and revising the work	Doing the work, repairing and revising, steering the loop, improving the system.	35.7%	1,016
Checking the output	Checking output for errors, calibrating reliance, judging fit to goal, verifying against outside evidence.	9.1%	260
Responsibility, social and self-directed work	Coordinating and deliberating with others, consent and disclosure, ethical judgement, communicating outward, holding accountability, care work; and learning, reflection and sustained attention to the system.	6.9%	197
Not stated†	The paper does not say what the person retains.	0.2%	7
What moves into the tool (n=2,326 papers)			
Computation and judgement	The tool predicts, simulates, fits models, calculates, optimises, estimates uncertainty, computes geometry; and attributes behaviour, classifies, detects errors, critiques, scores, infers a person's state, ranks and verifies.	30.5%	709
Representation and translation	The tool gives external form, converts between formats and languages, lays material out, summarises, verbalises, encodes data visually, coordinates views, compiles specifications; and finds groupings, patterns, similarities, aggregates and comparisons.	23.6%	549
Decision, guidance and action	The tool recommends, selects the next step or decides outright; supplies templates, procedures, decompositions, reference material, instruction and prompts; propagates judgments, does clerical work, operates environments, enforces constraints, fabricates; stands in for people, coordinates them, relays communication.	15.6%	364
Perception, search and memory	The tool senses, extracts, recognises, segments, transcribes and flags; finds and narrows material; records, stores, recalls and reminds; watches and tracks.	15.0%	349
Content generation	The tool generates new artifacts (alternatives, images, prose, code, 3D forms, audio) and responses (explanations, questions, answers, utterances, conversational replies, revisions of the person's material).	14.9%	346
Offloading that failed or was withheld	The offloading failed or was incomplete, or the tool deliberately stopped at a proposal and left the judgment with the person.	0.3%	8
Nothing offloaded, or not stated†	The value states that nothing moved to the tool, or names no operation at all.	0.0%	1

Table 11. The two dimensions about the work and the evidence: what happens to the work and how learning is evidenced, with the share and number of papers carrying each value. A dagger marks the absence of a value.

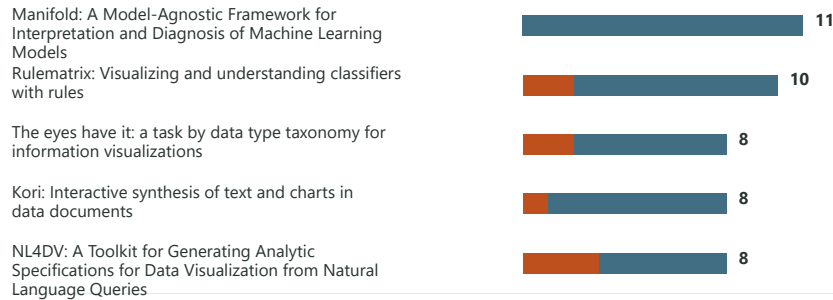
Dimension and value	Definition	Share	Papers
What happens to the work (n=2,643 papers)			
The person's work changes form	The person still works, differently: reading new representations, navigating and searching, steering and specifying at a higher level, reviewing and selecting instead of producing, checking and repairing, following a guided procedure, judging and reflecting on new grounds, coordinating through shared artifacts.	92.2%	2,437
A step becomes automatic	A step the person used to do is now done, or mostly done, by the system: automatic extraction or generation, exception-only attention, the same task made cheaper, automatic checking of the person's own work, effort freed for other work.	26.4%	697
Study condition or proposal, not practice	The value describes a study arm, protocol or manipulation, or a proposed or recommended change that was not observed in use.	16.1%	425
Work newly possible	The person can now do something they could not: work in a new setting or moment, at a larger scale or scope, or a task that was out of reach.	4.6%	121
New work created by the system	Labour that did not exist before the system: setting up, configuring and maintaining it, added steps or obligations, burden or unproductive activity it caused.	4.0%	105
Work unchanged	The practice did not change: the person keeps the core task and the system only informs it, under-uses or works around it, or nothing changed at all.	2.0%	54
How learning is evidenced (n=2,800 papers)			
A proxy measured instead	Capability stood in for by a proxy: subjective judgements about the system, usability instruments, qualitative accounts, attitudes or engagement; in-session task performance, behaviour or rated work products; or the quality of the system's own output, including the machine's learning.	37.6%	1,054
Learning measured	An objective capability measure was taken: immediately with support present, later or with support withdrawn, or with a null, mixed or adverse result, or compromised by confounding or deliberate suppression.	9.4%	263
Learning self-reported only	A learning or skill gain is asserted on self-report alone, or gestured at in remarks while the paper says learning itself was not evaluated.	4.6%	130
Learning not measured†	Nothing about the person's learning, transfer, retention or capability was measured: the value names no measure and no substitute, the authors say so themselves or defer it, there were no participants at all, or learning is the stated purpose yet never evaluated.	48.3%	1,353

Where prior work is used

1,642 distinct papers · 1,697 PDF-matched lineage cards

Most shared identifier-linked sources

Distinct papers with recorded use



Follow a source into later designs

Highest-ranked source with an unambiguous recorded year; every use is shown.

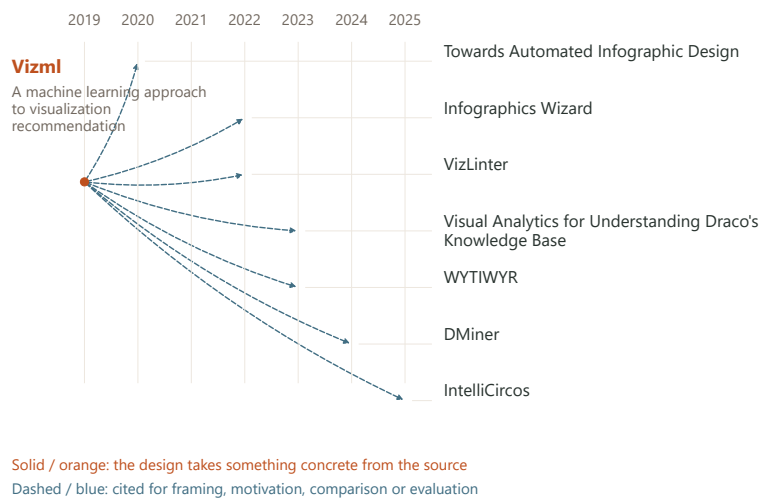


Fig. 14. **How papers use prior work.** Across 1,697 lineage cards for papers in the analysis, the upper panel ranks sources with a DOI or other identifier by the number of papers that use them. Orange marks papers whose design takes something concrete from the source (a mechanism, interface element, algorithm, representation or interaction); blue marks papers that cite it for framing, motivation, comparison or evaluation. Two language-model coders applied this one definition to every pair shown ($\kappa = 0.90$). Below, the top-ranked source with a known publication year connects to later designs, with solid arrows for the first kind of use and dashed arrows for the second. The lineage records link 43,733 source–paper pairs to passages and cards.

Shared sources across the two review streams

106,521 cited sources · 5,748 cited by five or more review papers

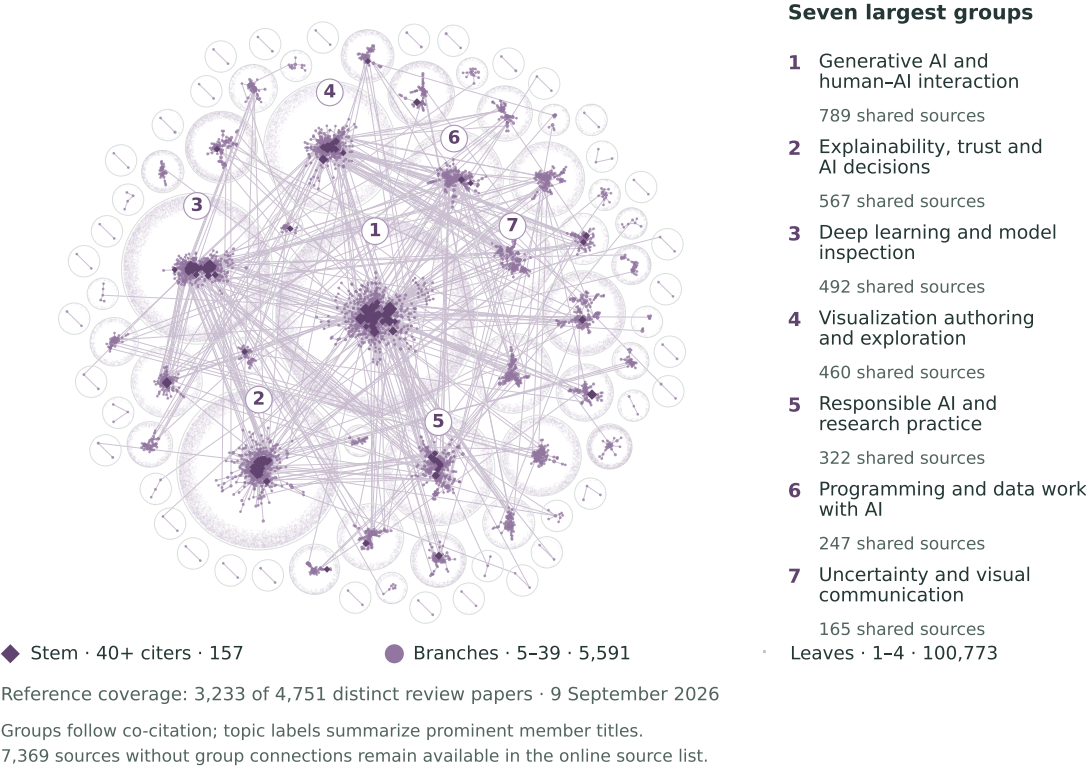


Fig. 15. **Shared-reference neighborhoods.** Reference metadata covers 3,233 of 4,751 papers, from bibliographic data of 9 September 2026, before the corpus was limited to AI papers. Numbers label the seven largest groups of sources cited together.

C DETAILED MECHANISM AND EVIDENCE PROFILES

Coding and populations. This appendix uses the coding of the mechanism cards completed by 28 August 2026, restricted to the systematic corpus (Section 4): 2,847 open-coded papers and 2,891 closed-field papers. Appendix A.6 lists the population behind each profile.

The complete facet catalogue supports the main mechanism analysis (Section 6); complementary evidence profiles support the workflow analysis (Section 5).

C.1 Category Maps and Mechanism Examples

The eighteen category maps cover all 110 mechanism facets in five families. Branches group design operations that systems may combine. Schematics explain the choices; source figures show cited implementations. Variations are recorded implementations, not settings available in every system. Map order and size carry no frequency or effectiveness scale.

Search the full definitions. Links beneath each map open its source mechanisms.

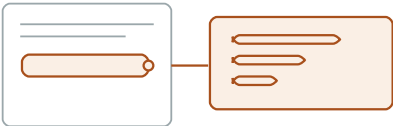
C.1.1 *Explanation and visibility.* What can a person inspect and understand?

FC-C01 / Explanation and visibility

Explaining system behaviour

Which aspect of the system is made inspectable?

Makes a system's outputs, behaviour, uncertainty or current state inspectable, so people can understand what it is doing and judge its results.



DESIGN CHOICES

RETAINED DISTINCTIONS

A particular result

—	FC001 output rationale exposed	FC002 feature attribution for an output
---	--	---

Process and sequence

—	FC006 live process-state visibility	FC008 stepwise reasoning or execution trace
---	---	---

Model and internals

—	FC003 global model-behaviour summary	FC007 internal or latent state surfaced
---	--	---

Reliability and limits

—	FC004 uncertainty and confidence exposed	FC005 capability and knowledge-boundary disclosure
---	--	--

Recorded form · FC001: Rationale for a specific machine output (VA001).

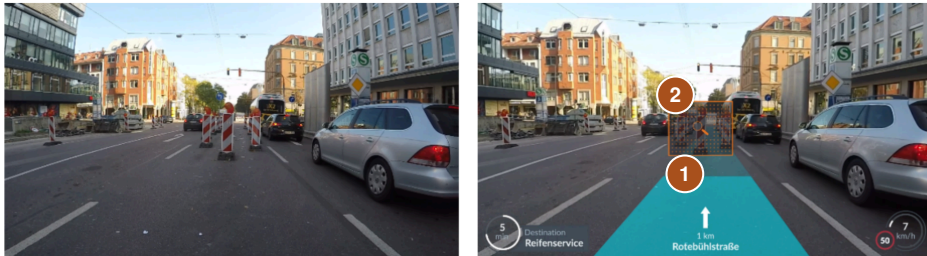
Recorded form · FC002: Feature or factor contribution to an outcome (VA002).

Fig. 16. Explaining system behaviour. Which aspect of the system is made inspectable? Separate the explanation of one result, the running process, the model across cases, and the limits of what it can establish. Category definitions; Browse category mechanisms. Recorded forms: Rationale for a specific machine output; Feature or factor contribution to an outcome.

FC-C01 / SOURCE ILLUSTRATION

Explaining system behaviour

PAPER ILLUSTRATION · FC006



1. Intended route overlaid on the road 2. Detected obstacle marked in the scene

Live overlays expose the vehicle's planned route and detected objects in the road scene.

ExplAIn Yourself! Transparency for Positive UX in Autonomous Driving - Figure 2, PDF page 4

Schneider et al.; CC BY; cropped; numbered annotations added by this review.

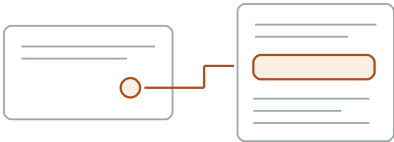
Fig. 17. Explaining system behaviour: source illustration. Figure 2, PDF page 4 in [120]. Live overlays expose the vehicle's planned route and detected objects in the road scene. Inspect the illustrated mechanism.

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FC-C02 / Explanation and visibility
Sources and evidence

How can a result be traced or checked?

Connects a result to its sources, supporting evidence or underlying cases, so people can trace where it came from and check its basis.



DESIGN CHOICES

RETAINED DISTINCTIONS

Origin and derivation

FC009

output-to-source provenance link

FC012

analytic-history provenance capture

Underlying cases

FC010

aggregate-to-instance drill-down

Evidence for a claim

FC011

evidence presented for verification

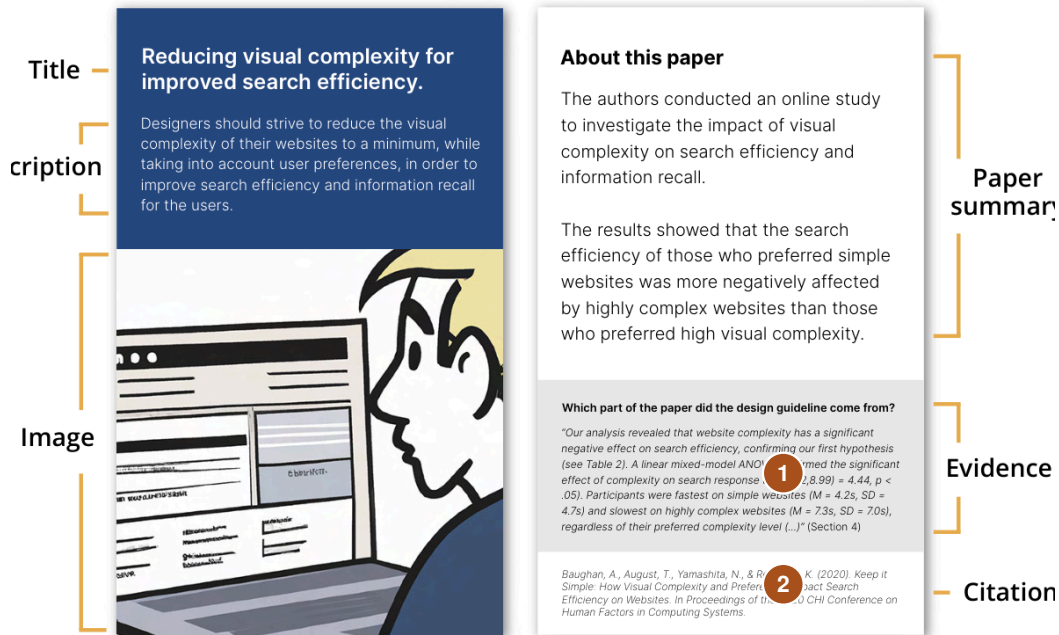
Recorded form · FC010: Drill-down from summary to underlying instances (VA012).

Fig. 18. Sources and evidence. How can a result be traced or checked? Tracing origin and derivation, opening underlying cases, and assessing evidence answer different checking questions. Category definitions; Browse category mechanisms. Recorded forms: Drill-down from summary to underlying instances.

FC-C02 / SOURCE ILLUSTRATION

Sources and evidence

PAPER ILLUSTRATION · FC009



1. Supporting passage 2. Citation back to the paper

The design card places a supporting source passage and its citation beside the implication.

From Paper to Card: Transforming Design Implications with Generative AI · Figure 1, PDF page 2

Shin et al.; CC BY; cropped; numbered annotations added by this review.

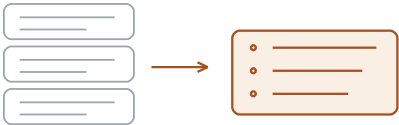
Fig. 19. Sources and evidence: source illustration. Figure 1, PDF page 2 in [125]. The design card places a supporting source passage and its citation beside the implication. Inspect the illustrated mechanism.

FC-C03 / Explanation and visibility

Summarising and labelling content

How is content condensed, interpreted or recast?

Condenses or interprets content through summaries, labels, profiles or a change of representation, so people can grasp what it means.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Condense content	FC107 summarisation and condensation	
Characterise an item	FC105 machine-generated semantic labelling	FC109 entity or concept profile
Change its form	FC108 semantic transformation into another form	

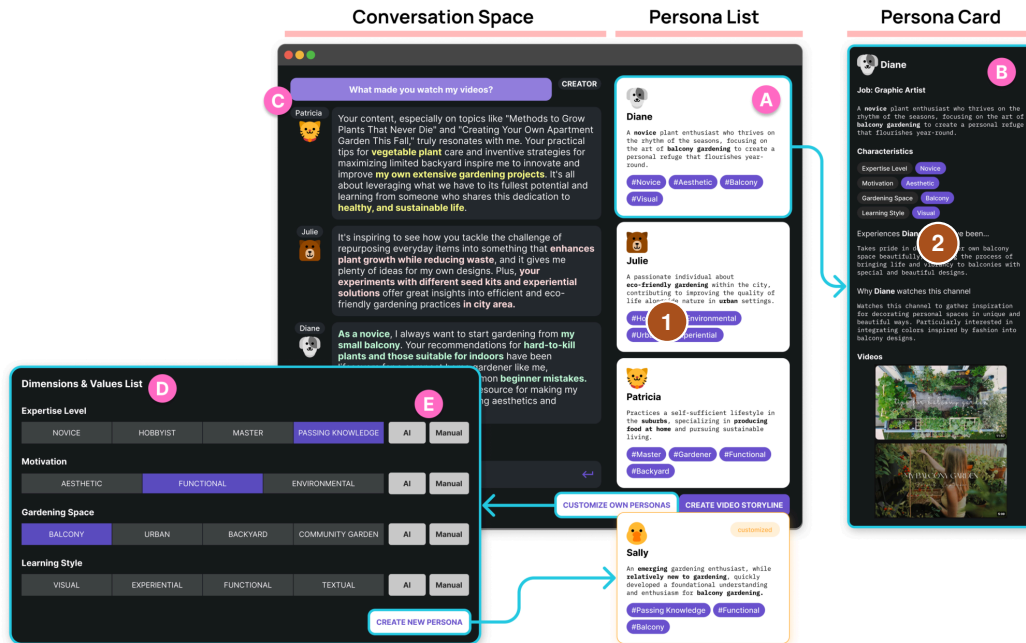
Recorded form · FC107: Condensed summary of a single item (VA097).

Fig. 20. Summarising and labelling content. How is content condensed, interpreted or recast? Distinguish reducing a body of content, characterising an item, and changing representational kind. Category definitions; Browse category mechanisms. Recorded forms: Condensed summary of a single item.

FC-C03 / SOURCE ILLUSTRATION

Summarising and labelling content

PAPER ILLUSTRATION · FC109



1. Compact audience profiles 2. Expanded persona details

Audience segments become profile cards, with detailed experiences and motivations available on selection.

Proxona: Supporting Creators' Sensemaking and Ideation with LLM-Powered Audience Personas · Figure 2, PDF page 6

Choi et al.; CC BY; cropped; numbered annotations added by this review.

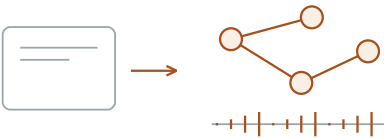
Fig. 21. Summarising and labelling content: source illustration. Figure 2, PDF page 6 in [30]. Audience segments become profile cards, with detailed experiences and motivations available on selection. Inspect the illustrated mechanism.

FC-C04 / Explanation and visibility

Visual and sensory representation

Through which form can a person perceive meaning?

Expresses meaning through visual form, language, sound, touch or physical action, including translations and adaptations that make it accessible.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Perceptual carrier	FC085 visual encoding and graphical mapping	FC083 non-visual sensory encoding	FC086 embodied or physical enactment
Translation of meaning	FC084 cross-modal translation		FC089 verbal description of perceptual content
Adaptation and reframing	FC088 accessible or simplified presentation		FC087 metaphorical or symbolic re-representation

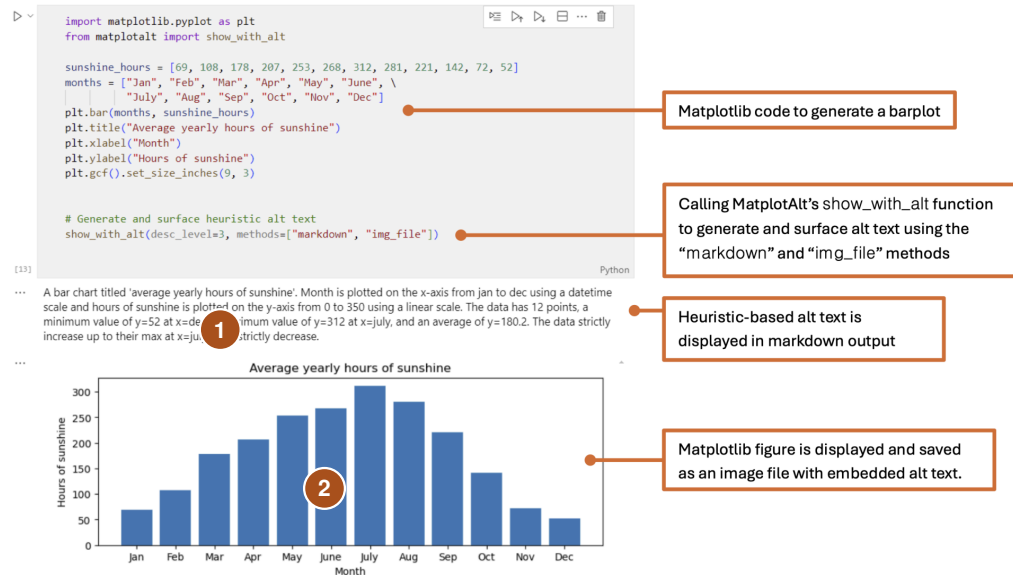
Recorded form · FC083: Spoken or audio rendering of content (VA071).

Fig. 22. Visual and sensory representation. Through which form can a person perceive meaning? Separate the perceptual carrier, translation between carriers, and adaptations that change how the meaning can be grasped. Category definitions; Browse category mechanisms. Recorded forms: Spoken or audio rendering of content.

FC-C04 / SOURCE ILLUSTRATION

Visual and sensory representation

PAPER ILLUSTRATION · FC088



1. Description surfaced as text 2. The same chart remains visible

A notebook presents a chart and its generated description together; the library also supports alternative output placements.

MatplotlibAlt: A Python Library for Adding Alt Text to Matplotlib Figures in Computational Notebooks · Figure 1, PDF page 1

Nylund et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 23. Visual and sensory representation: source illustration. Figure 1, PDF page 1 in [102]. A notebook presents a chart and its generated description together; the library also supports alternative output placements. Inspect the illustrated mechanism.

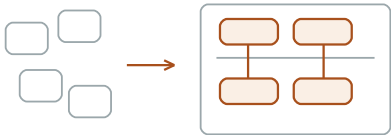
C.1.2 Organisation and comparison. How can a person organise and compare material?

FC-C05 / Organisation and comparison

Organising and structuring

Which relationships give material its structure?

Arranges information into relationships, groups, levels or sequences, so people can understand its structure and move between the whole and its parts.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Links and containment	FC015 relational or network map	FC013 hierarchical or nested organisation	
Groups and categories	FC016 similarity grouping and clustering	FC017 categorisation scheme or controlled vocabulary	FC106 human semantic coding and categorisation
Whole, parts and detail	FC014 overview or aggregate gist	FC018 decomposition into parts	FC019 multiscale detail and semantic zoom
Position and alignment	FC020 temporal arrangement of material	FC021 spatial arrangement of material	FC022 cross-representation correspondence
Quantitative structure	FC023 distributional or statistical structure display		

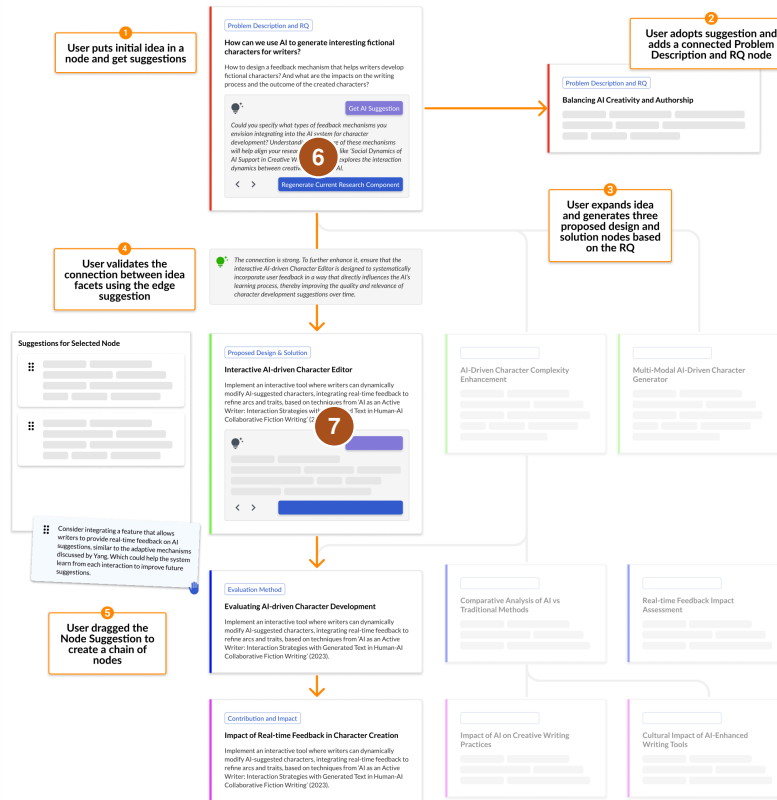
Recorded form - FC015: Relational graph or dependency network (VA019).

Fig. 24. Organising and structuring. Which relationships give material its structure? Relations, category membership, granularity, position and statistical structure organise different aspects of the same material and may be combined. Category definitions; Browse category mechanisms. Recorded forms: Relational graph or dependency network.

FC-C05 / SOURCE ILLUSTRATION

Organising and structuring

PAPER ILLUSTRATION · FC018



6. Research question node 7. Linked solution node

An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes.

Iterative Research Idea Development Through Evolving and Composing Idea Facets with Literature-Grounded Feedback · Figure 4, PDF page 8

Pu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 25. Organising and structuring: source illustration. Figure 4, PDF page 8 in [110]. An idea is decomposed into linked, editable research-question, solution, evaluation and contribution nodes. Inspect the illustrated mechanism.

FC-C06 / Organisation and comparison

Comparing alternatives

What is placed in a common comparison frame?

Places alternatives, cases or states in a common frame so people can examine their similarities, differences and trade-offs.

	A	B

DESIGN CHOICES

RETAINED DISTINCTIONS

Separate candidates

FC024
side-by-side alternative comparison

Changed states or versions

FC025
contrastive or counterfactual comparison

FC028
change-over-time comparison

Standards and populations

FC026
benchmark or peer-referenced comparison

FC027
subgroup or population comparison

Recorded form · FC024: Side-by-side comparison of alternatives (VA023).

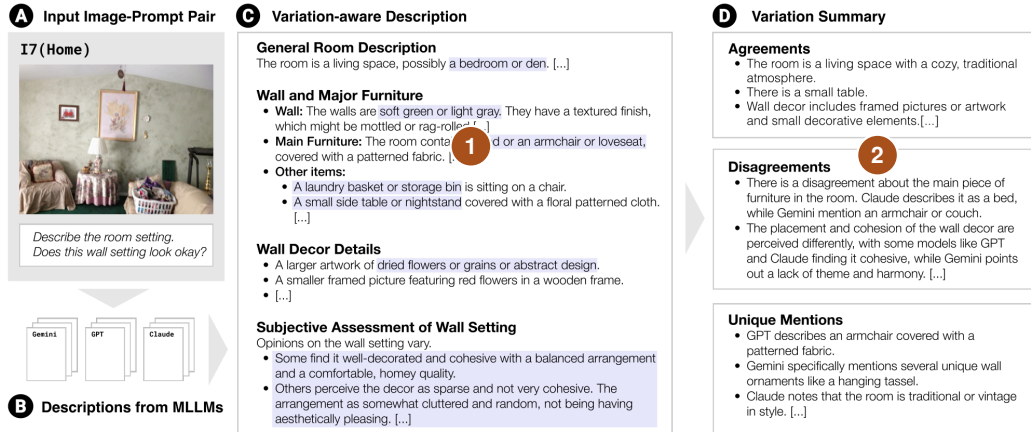
Recorded form · FC025: Before-and-after or version comparison (VA025).

Fig. 26. Comparing alternatives. What is placed in a common comparison frame? The comparand distinguishes candidate selection, variation of the same object, and comparison against groups or reference points. Category definitions; Browse category mechanisms. Recorded forms: Side-by-side comparison of alternatives; Before-and-after or version comparison.

FC-C06 / SOURCE ILLUSTRATION

Comparing alternatives

PAPER ILLUSTRATION · FC024



1. Detailed merged description 2. Agreements and disagreements

The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary.

Surfacing Variations to Calibrate Perceived Reliability of MLLM-generated Image Descriptions · Figure 2, PDF page 5

Chen et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 27. Comparing alternatives: source illustration. Figure 2, PDF page 5 in [26]. The interface aligns multiple image descriptions in a detailed narrative and a separate discrepancy summary. Inspect the illustrated mechanism.

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FC-C07 / Organisation and comparison
Search, filtering and ranking

How is relevant material found and brought into view?

Helps people find relevant material and control what is in view through search, navigation, filters, rankings and scope choices.

The diagram illustrates a search interface. At the top is a search bar with a magnifying glass icon on the left and a text input field. Below the search bar are three filter buttons, each with a text input field and a dropdown arrow on the right. The top filter button is highlighted with an orange border.

DESIGN CHOICES

RETAINED DISTINCTIONS

Define the request

FC030
query construction and search specification

FC033
analytic scope setting

Select and order

FC029
relevance filtering of a corpus

FC032
ranked or prioritised surfacing

Navigate and look up

FC034
navigation and wayfinding through structure

FC031
on-demand lookup of related information

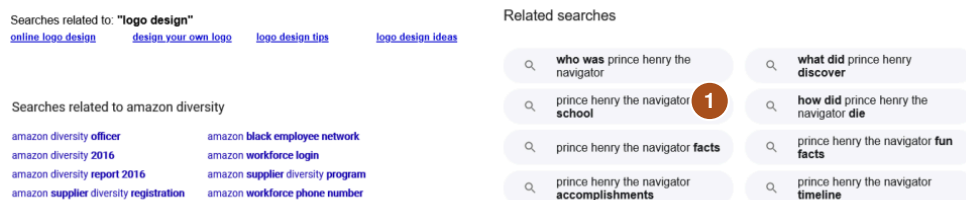
Recorded form · FC030: Filter, query or search over a set (VA028).

Fig. 28. Search, filtering and ranking. How is relevant material found and brought into view? Defining the request, selecting or ordering results, and moving through material are distinct retrieval operations. Category definitions; Browse category mechanisms. Recorded forms: Filter, query or search over a set.

FC-C07 / SOURCE ILLUSTRATION

Search, filtering and ranking

PAPER ILLUSTRATION · FC030



1. Follow-on query formulations

Related-search suggestions provide alternative formulations for continuing a search.

From 10 Blue Links Pages to Feature-Full Search Engine Results Pages - Analysis of the Temporal Evolution of SERP Features · Figure 16, PDF page 5

Oliveira et al.; CC BY; cropped; numbered annotations added by this review.

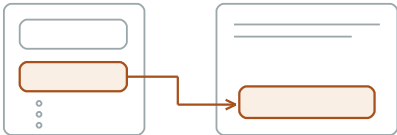
Fig. 29. Search, filtering and ranking: source illustration. Figure 16, PDF page 5 in [103]. Related-search suggestions provide alternative formulations for continuing a search. Inspect the illustrated mechanism.

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FC-C08 / Organisation and comparison
Notes, history and memory

What is kept available for continuing the work?

Keeps information and work history available through notes, annotations, memory aids and replay, so people can revisit and continue their work.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Express and capture	FC077 externalising internal structure into an artifact	FC080 capture of activity as it happens
Mark and arrange	FC078 annotation onto material	FC082 reorganisable working workspace
Recall and revisit	FC079 external memory and recall cue	FC081 replay and history navigation

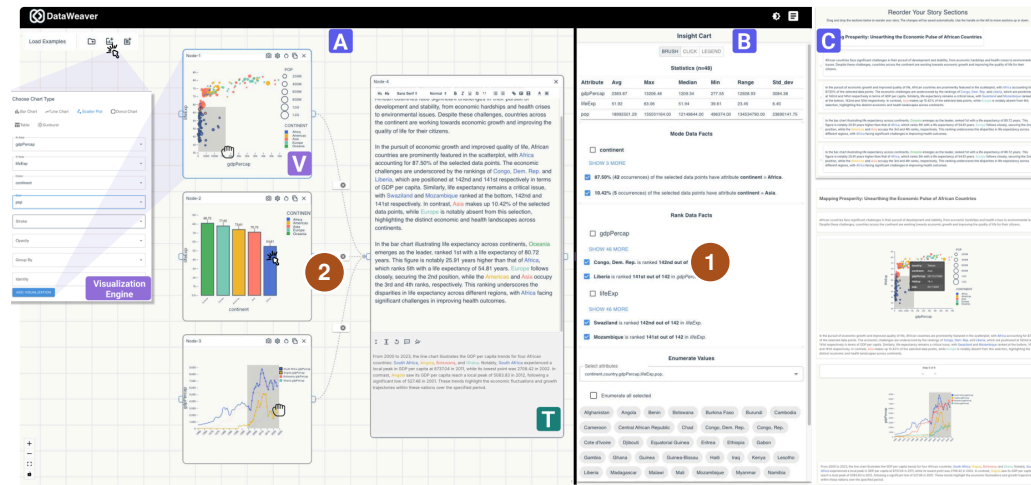
Recorded form · FC078: Annotation and comment capture on content (VA048).

Fig. 30. Notes, history and memory. What is kept available for continuing the work? Separate making or capturing a record, arranging material at hand, and returning to an earlier state. Category definitions; Browse category mechanisms. Recorded forms: Annotation and comment capture on content.

FC-C08 / SOURCE ILLUSTRATION

Notes, history and memory

PAPER ILLUSTRATION · FC082



1. Facts held in the Insight Cart 2. Connected story components

An Insight Cart holds candidate facts for later selection into a connected data story.

DATAWEAVER: Authoring Data-Driven Narratives through the Integrated Composition of Visualization and Text · Figure 2, PDF page 5

Fu et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 31. Notes, history and memory: source illustration. Figure 2, PDF page 5 in [41]. An Insight Cart holds candidate facts for later selection into a connected data story. Inspect the illustrated mechanism.

C.1.3 User control and expression. What can a person express, change or try?

FC-C09 / User control and expression

Goals and constraints

How does a person specify what the system should do?

Lets people state what they want through goals, rules, criteria, examples or structured inputs that the system can act on.

DESIGN CHOICES

RETAINED DISTINCTIONS

Words and demonstrations

FC053
natural-language or conversational specification

FC054
example or demonstration specification

Structured and executable input

FC057
structured input form or schema

FC058
authoring a formal or executable representation

Conditions and criteria

FC055
explicit constraint or rule specification

FC056
criteria and rubric specification

Recorded form · FC055: Rule, criterion or rubric authoring (VA042).

Fig. 32. Goals and constraints. How does a person specify what the system should do? Distinguish conveying intent, providing a structured or executable specification, and declaring constraints or evaluation criteria. Category definitions; Browse category mechanisms. Recorded forms: Rule, criterion or rubric authoring.

FC-C09 / SOURCE ILLUSTRATION

Goals and constraints

PAPER ILLUSTRATION · FC054

☐ Show only Comments labeled as "Remove" 30 / 30 labeled examples; 14 example(s) labeled as "Remove".

Those people who pull out guns just because they get upset are called gang banger thug losers. 99% of their guns are illegally possessed. Legal gun owners aren't taking out their guns during an argument at parties.

Remove **1** Keep

So I should get rid of my gun because some idiot killed.

Remove Keep

Idiot sandwich 🥪

Remove Keep

Load More Examples Create **2** your Filter

1. Keep or remove this example 2. Train the filter from labels

Situated keep/remove judgments supply examples of the content a person wants the classifier to retain.

End User Authoring of Personalized Content Classifiers: Comparing Example Labeling, Rule Writing, and LLM Prompting · Figure 2, PDF page 6

Wang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 33. Goals and constraints: source illustration. Figure 2, PDF page 6 in [137]. Situated keep/remove judgments supply examples of the content a person wants the classifier to retain. Inspect the illustrated mechanism.

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FC-C10 / User control and expression

Editing and steering

Where can a person intervene in the work?

Lets people change a representation, revise an output or redirect a process through edits, settings, parameters and reversible actions.



DESIGN CHOICES

RETAINED DISTINCTIONS

Object and output

FC047

direct manipulation of a semantic object

FC048

editable machine output

Settings and quantities

FC050

configuration of system behaviour

FC049

continuous parameter or dial control

Direction and commitment

FC051

steering an ongoing process

FC052

reversibility and safe experimentation

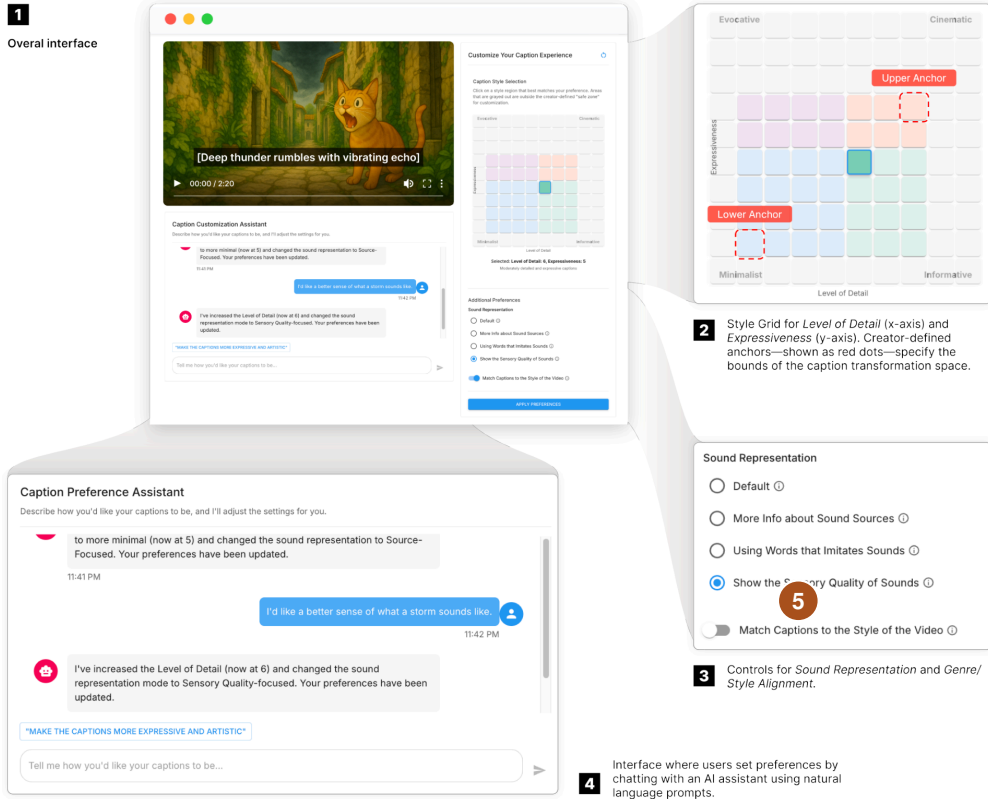
Recorded form · FC049: Parameter, threshold or weight controls (VA039).

Fig. 34. Editing and steering. Where can a person intervene in the work? Editing an object, configuring behaviour, steering the current process and controlling commitment operate at different points. Category definitions; Browse category mechanisms. Recorded forms: Parameter, threshold or weight controls.

FC-C10 / SOURCE ILLUSTRATION

Editing and steering

PAPER ILLUSTRATION · FC050



CapTune: Adapting Non-Speech Captions With Anchored Generative Models · Figure 4, PDF page 8

Huang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 35. Editing and steering: source illustration. Figure 4, PDF page 8 in [56]. Viewers switch between sound sources, imitations and sensory descriptions for the same captioned material. Inspect the illustrated mechanism.

FC-C11 / User control and expression

Generating and choosing alternatives

How are possibilities offered and shaped?

Provides candidate outputs, continuations or a space of possibilities that people can explore, constrain and select from.



DESIGN CHOICES	RETAINED DISTINCTIONS	
Present a choice space	FC035 candidate or option set to choose from	FC038 option-space landscape or design space
Generate or continue	FC036 generated alternatives to explore	FC110 completion and continuation of content
Bound generation	FC037 user-constrained generation	

Recorded form · FC036: Gallery of generated alternatives to choose among (VA033).
Recorded form · FC110: Completion or continuation of what the person began (VA036).

Fig. 36. Generating and choosing alternatives. How are possibilities offered and shaped? Separate presenting a choice space, producing new content, and bounding the generation process. A continuation is not necessarily a set of alternatives. Category definitions; Browse category mechanisms. Recorded forms: Gallery of generated alternatives to choose among; Completion or continuation of what the person began.

FC-C11 / SOURCE ILLUSTRATION

Generating and choosing alternatives

PAPER ILLUSTRATION · FC110

Double-click the snippet to edit it directly. **THIS IS SUGGESTED TEXT**

Adjust role

Explain interaction × ⓘ

Adjust formatting

heading drophead body **note**

Get recommendations

Highlight interactive elements and guide the audience's interaction flow using moderately complex sentences. Use imperative sentences to instruct users on actions they can perform. Depending on the complexity, aim for 10-25 words, a readability level appropriate for someone with a 6th to 10th grade education, and a word variety of 50-70%.

Word count **5** ⓘ

Word variety **40%** ⓘ

Readability score **69** ⓘ

Your text is likely to be understood by someone who has at least a 8th-grade education.

Generate text

Re-generate Shorten **1** Simplify

Add context to incorporate into generation

What should we consider when generating this text?

Delete snippet

1. Regenerate, shorten or simplify

A text snippet exposes explicit actions for generating a revised version while the author continues composing.

Plume: Scaffolding Text Composition in Dashboards · Figure 5, PDF page 11

Lisnic et al.; CC BY; cropped; numbered annotations added by this review.

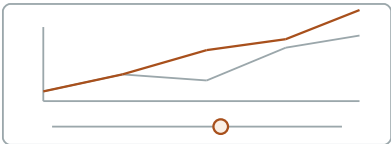
Fig. 37. Generating and choosing alternatives: source illustration. Figure 5, PDF page 11 in [79]. A text snippet exposes explicit actions for generating a revised version while the author continues composing. Inspect the illustrated mechanism.

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FC-C12 / User control and expression
What-if exploration and rehearsal

What can be tried before acting in the real setting?

Lets people try hypothetical changes, probe a system's behaviour or practise in a simulated setting and inspect what happens.



DESIGN CHOICES	RETAINED DISTINCTIONS
Probe the system	FC098 system-behaviour probing
Construct a scenario	FC097 scenario construction and what-if exploration
Rehearse an activity	FC096 simulated environment for rehearsal

Recorded form · FC096: Practice, rehearsal or simulation environment (VA058).

Fig. 38. What-if exploration and rehearsal. What can be tried before acting in the real setting? Separate learning how a system responds, constructing a hypothetical world, and rehearsing a person's activity. Category definitions; Browse category mechanisms. Recorded forms: Practice, rehearsal or simulation environment.

FC-C12 / SOURCE ILLUSTRATION

What-if exploration and rehearsal

PAPER ILLUSTRATION · FC096



1. Dialogue history 2. Teacher question entry

Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history.

Pattern analysis of ambitious science talk between preservice teachers and AI-powered student agents · Figure 1, PDF page 3

Barrett et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 39. What-if exploration and rehearsal: source illustration. Figure 1, PDF page 3 in [9]. Preservice teachers rehearse classroom discussion with represented students and a persistent dialogue history. Inspect the illustrated mechanism.

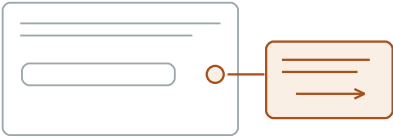
C.1.4 *Guidance and feedback.* How is the next action or reflection supported?

FC-C13 / Guidance and feedback

Recommendations and guidance

What helps a person decide where to look or what to do?

Directs attention or suggests an action, resource or procedure, including corrective advice, consequence previews and warnings.



DESIGN CHOICES

RETAINED DISTINCTIONS

Notice and hold back

FC043
attention direction and salience
cueing

FC044
risk alert or warning

**Preview a
consequence**

FC046
prospective consequence preview

**Choose an action or
resource**

FC040
next-step recommendation

FC042
item or content recommendation

Perform or repair

FC045
procedural how-to instruction

FC041
corrective or remediation guidance

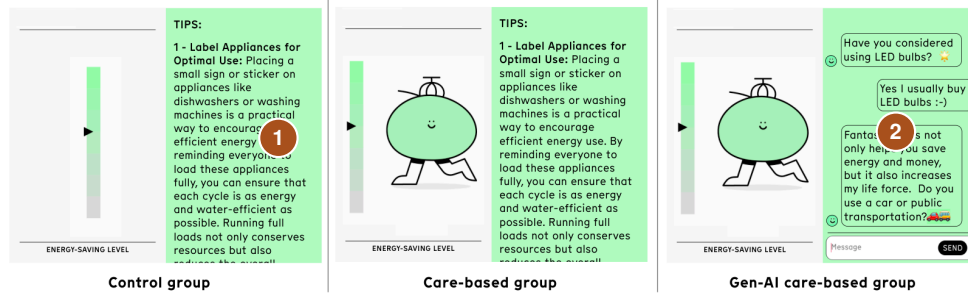
Recorded form · FC046: Preview of the consequence before committing (VA037).

Fig. 40. Recommendations and guidance. What helps a person decide where to look or what to do? Attention and warnings, previews, recommendations, and procedural or corrective guidance contribute different information to an action. Category definitions; Browse category mechanisms. Recorded forms: Preview of the consequence before committing.

FC-C13 / SOURCE ILLUSTRATION

Recommendations and guidance

PAPER ILLUSTRATION · FC040



1. Fixed tips 2. Conversational condition

The study contrasts fixed energy-saving tips with a conversational condition that elicits household context.

Care-Based Eco-Feedback Augmented with Generative AI: Fostering Pro-Environmental Behavior through Emotional Attachment · Figure 6, PDF page 6

Berney et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 41. Recommendations and guidance: source illustration. Figure 6, PDF page 6 in [12]. The study contrasts fixed energy-saving tips with a conversational condition that elicits household context. Inspect the illustrated mechanism.

FC-C14 / Guidance and feedback

Feedback and checking

What basis is provided for judging the work?

Shows how work is progressing or how a result meets a goal, criterion or reference. It includes feedback on attempts, problem diagnosis, quality assessments and checks.



DESIGN CHOICES

RETAINED DISTINCTIONS

Attempt and progress

FC065
formative outcome feedback

FC066
progress and status tracking

Problem, quality and status

FC063
Identifying problems in the work

FC064
evaluative judgment or quality rating

FC067
correctness or status marking

Check against a reference

FC068
verification and checking against a reference

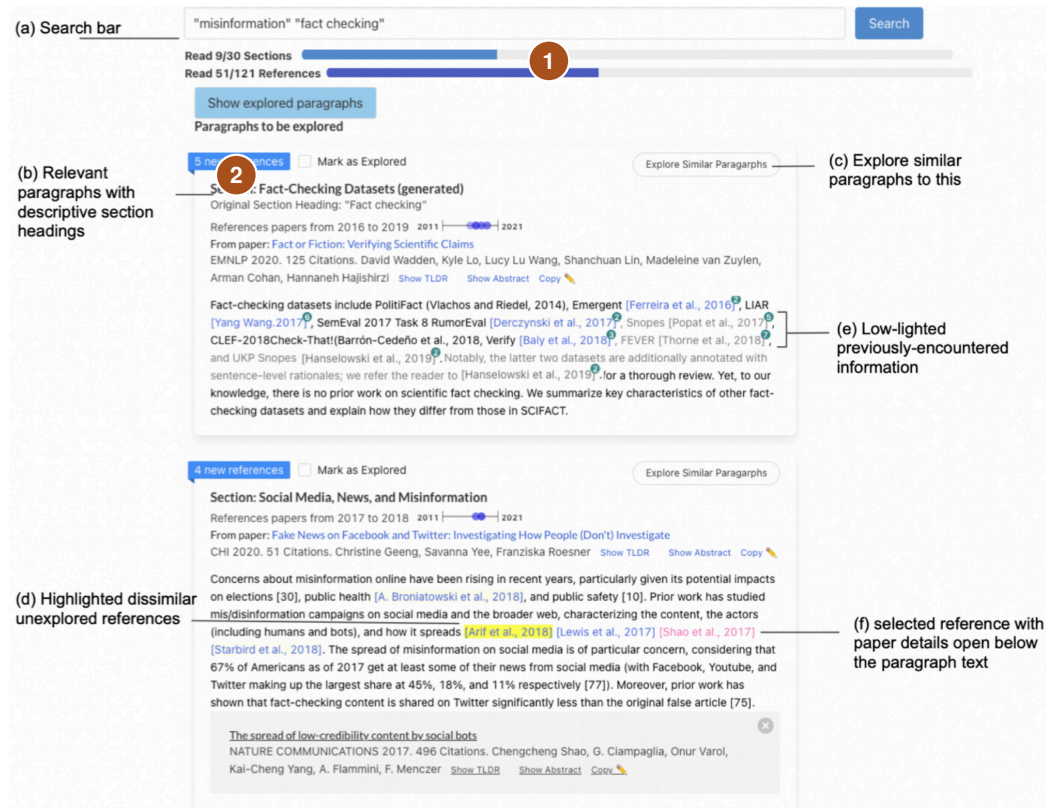
Recorded form · FC063: Error and failure diagnosis (VA007).

Fig. 42. Feedback and checking. What basis is provided for judging the work? Separate feedback about an attempt or progress, appraisals and status marks, and an active check against a reference. Category definitions; Browse category mechanisms. Recorded forms: Error and failure diagnosis.

FC-C14 / SOURCE ILLUSTRATION

Feedback and checking

PAPER ILLUSTRATION · FC066



1. Paragraph and reference coverage 2. Unread-reference count

Persistent coverage bars and local unread-reference counts make progress through the literature visible.

Relatedly: Scaffolding Literature Reviews with Existing Related Work Sections · Figure 1, PDF page 2

Palani et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 43. Feedback and checking: source illustration. Figure 1, PDF page 2 in [105]. Persistent coverage bars and local unread-reference counts make progress through the literature visible. Inspect the illustrated mechanism.

FC-C15 / Guidance and feedback

Instruction and paced help

What help is available, and when is it available?

Provides explanations, examples and orientation, and controls when or how much help is available through staged disclosure, learning paths or changing support.



DESIGN CHOICES

RETAINED DISTINCTIONS

Explain, exemplify and orient

FC101

conceptual explanation of domain meaning

FC102

worked example or exemplar

FC103

orientation and onboarding

Change amount or release

FC099

graduated or faded support

FC100

progressive disclosure of detail

FC039

deliberate restriction or withdrawal of support

Sequence learning sessions

FC104

curriculum and learning-path structuring

Recorded form · FC099: Graduated help that escalates or fades by response (VA093).

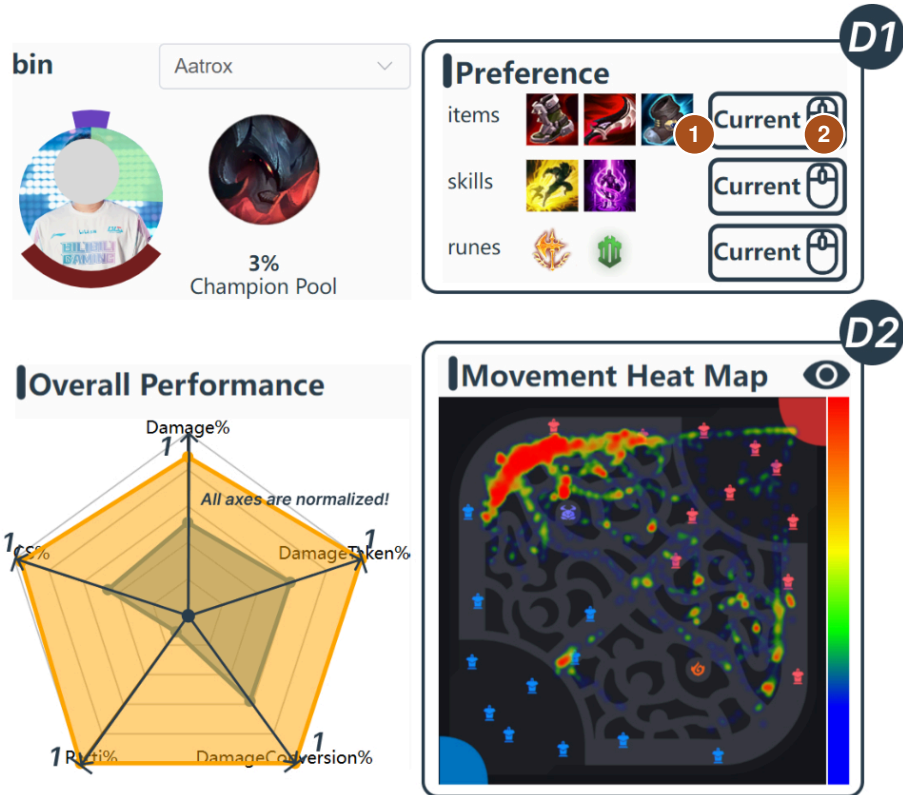
Recorded form · FC102: Worked or contrastive examples as instruction (VA057).

Fig. 44. Instruction and paced help. What help is available, and when is it available? Instructional content, the amount or timing of support, and progression across sessions are distinct design choices. Fading is not automatically evidence of learning. Category definitions; Browse category mechanisms. Recorded forms: Graduated help that escalates or fades by response; Worked or contrastive examples as instruction.

FC-C15 / SOURCE ILLUSTRATION

Instruction and paced help

PAPER ILLUSTRATION · FC102



1. Historical equipment choices 2. Switch to current choices

Historical equipment, skill and rune choices can be inspected beside controls revealing the current match.

StratIncon Detector: Analyzing Strategy Inconsistencies Between Real-Time Strategy and Preferred Professional Strategy in MOBA Esports · Figure 8, PDF page 12

Ma et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 45. Instruction and paced help: source illustration. Figure 8, PDF page 12 in [84]. Historical equipment, skill and rune choices can be inspected beside controls revealing the current match. Inspect the illustrated mechanism.

FC-C16 / Guidance and feedback

Structuring the workflow

How are stages, responsibilities and continuity arranged?

Organises work into stages, loops, roles and decision points, with ways to coordinate actions, hand work over, recover from problems and resume later.



DESIGN CHOICES	RETAINED DISTINCTIONS		
Sequence and recurrence	—	FC069 staged or phased workflow	FC070 iterative revision loop
Responsibility and routing	—	FC071 human decision checkpoint or gate	FC072 division of labour across agents
			FC074 orchestration and routing of actions
Continuity and recovery	—	FC073 handoff of work state	FC075 interaction repair and recovery
			FC076 resumable and persistent work state

Recorded form · FC075: System move that clarifies, confirms or repairs (VA038).

Fig. 46. Structuring the workflow. How are stages, responsibilities and continuity arranged? Separate progression through work, allocation of actions and authority, and continuity across interruptions or handoffs. Category definitions; Browse category mechanisms. Recorded forms: System move that clarifies, confirms or repairs.

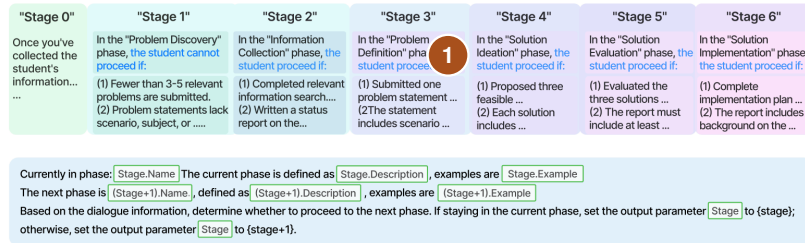
FC-C16 / SOURCE ILLUSTRATION

Structuring the workflow

PAPER ILLUSTRATION · FC074

Controller Agent Prompts

[Stage Decision Prompt]



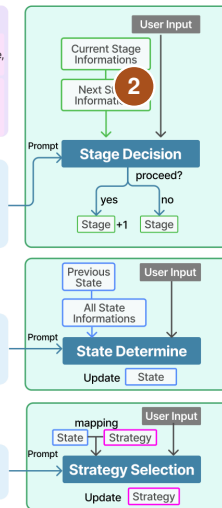
[State Determine Prompt]

If not moving to the next phase, the student's status may be: `State.Name` `State.Description`
 The student's previous status list is `[ State ]`. Remove all '0' and '1', and based on the new dialogue record, update and output the student's status list state.
 If moving to the next phase, the student's status will be "start," and update and output the student's status list as `state=[0]`.

[Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: `State.Name` `State.StrategyMapping` `Strategy.name` `Strategy.Description`
 Based on the dialogue information, output the most urgent problem ID as `urge_state` and the corresponding best strategy ID as `strategy`

Controller Agent Pipeline



1. Stage-specific completion criteria 2. Decision to advance or remain

Stage criteria and a controller decision determine whether the creative problem-solving workflow advances.

Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process · Figure 2, PDF page 9

Zha et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 47. Structuring the workflow: source illustration. Figure 2, PDF page 9 in [149]. Stage criteria and a controller decision determine whether the creative problem-solving workflow advances. Inspect the illustrated mechanism.

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FC-C17 / Guidance and feedback
Questions and reflection prompts

Which part of a person's thinking is elicited?

Asks people to express their goals, questions, explanations or reflections, making their own reasoning available to the work.

?

DESIGN CHOICES

RETAINED DISTINCTIONS

Needs and intentions



FC059
intent and need elicitation

Account and reasoning



FC062
narrative and reasoning elicitation

Questions and hypotheses



FC061
inquiry and hypothesis prompting

Reflection on thinking



FC060
reflective and metacognitive prompting

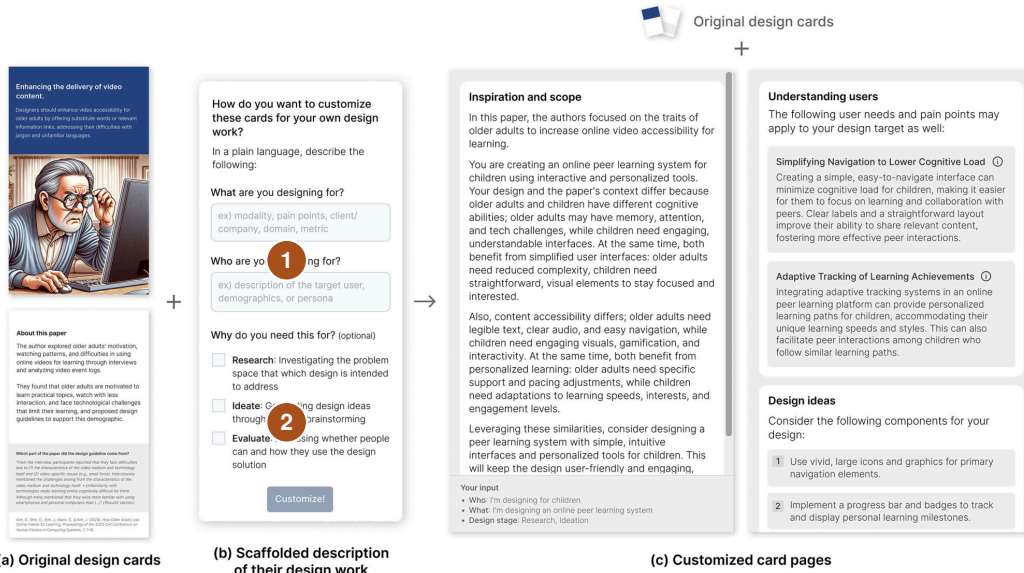
Recorded form · FC060: Reflection or self-assessment prompt (VA056).

Fig. 48. Questions and reflection prompts. Which part of a person’s thinking is elicited? The object of the prompt distinguishes needs, an account of reasoning, inquiry about the material, and reflection on one’s own thinking. Category definitions; Browse category mechanisms. Recorded forms: Reflection or self-assessment prompt.

FC-C17 / SOURCE ILLUSTRATION

Questions and reflection prompts

PAPER ILLUSTRATION · FC059



1. Who and what is being designed 2. Select the design stage

A guided form elicits design context before the system customizes research cards.

What About My Design Context?: Exploring the Use of Generative AI to Support Customization of Translational Research Artifacts · Figure 1, PDF page 7

Shin et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 49. Questions and reflection prompts: source illustration. Figure 1, PDF page 7 in [124]. A guided form elicits design context before the system customizes research cards. Inspect the illustrated mechanism.

C.1.5 Collaboration and perspective. How can people share work and perspectives?

FC-C18 / Collaboration and perspective

Shared views and perspectives

How do contributions and viewpoints become shared?

Makes people's contributions and viewpoints available to one another through shared representations, structured discussion, role framing and control over what is disclosed.



DESIGN CHOICES

RETAINED DISTINCTIONS

Shared work and awareness

FC090
shared co-constructed representation

FC091
visibility of others' state and judgment

Viewpoints and exchange

FC092
perspective-taking and viewpoint framing

FC094
role or persona framing of the interaction

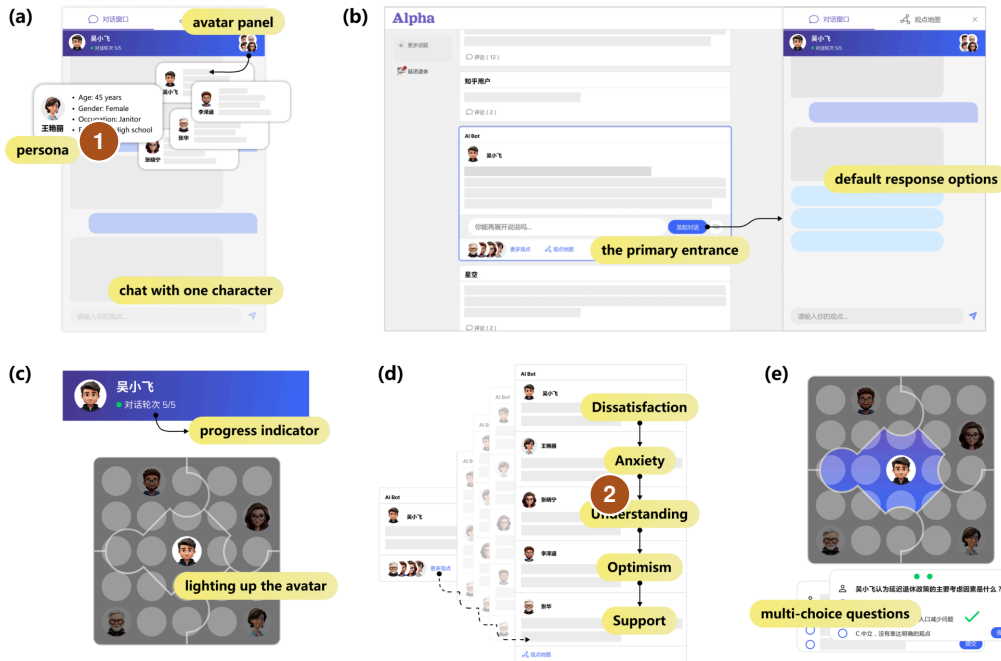
FC093
structured deliberation and perspective exchange

Control of disclosure

FC095
disclosure and privacy control

Recorded form - FC090: Shared workspace for joint work (VA065).

Fig. 50. Shared views and perspectives. How do contributions and viewpoints become shared? Separate shared artifacts and awareness, the framing and exchange of perspectives, and control of disclosure. Category definitions; Browse category mechanisms. Recorded forms: Shared workspace for joint work.

FC-C18 / SOURCE ILLUSTRATION**Shared views and perspectives****PAPER ILLUSTRATION · FC092**

1. Distinct stakeholder profiles 2. Progressive viewpoint sequence

Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure.

See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles · Figure 3, PDF page 7

Zhang et al.; CC BY; cropped; numbered annotations added by this review.

Fig. 51. Shared views and perspectives: source illustration. Figure 3, PDF page 7 in [152]. Personas expose distinct stakeholder positions; separate controls support dialogue and staged exposure. Inspect the illustrated mechanism.

C.2 Complementary Evidence Profiles

The learning-by-family matrix uses the open-code classification and its 2,761-paper matched population. It is distinct from the closed-field learning count of 301 among 2,891 included papers.

Claim support and developmental risk indicators

Each row is a mutually exclusive coding over the same 2,891 included cards.

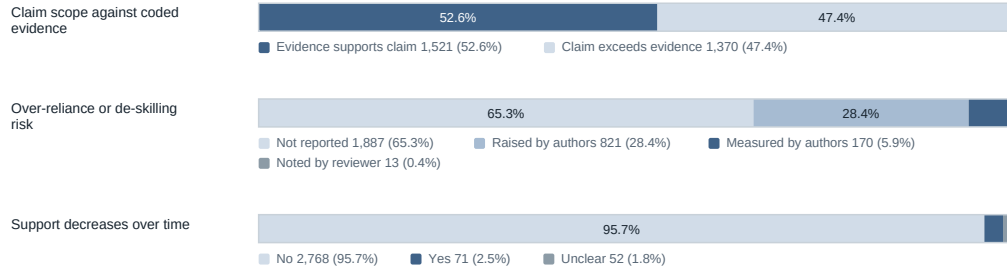


Fig. 52. **Claim scope exceeds coded evidence in nearly half of included cards, over-reliance is raised far more often than it is measured, and decreasing support is rare.** Each row is a mutually exclusive coding over the same 2,891 included cards, and every segment prints its own count and share. A claim-evidence gap records the distance between a card's stated claim and its linked evidence; it is not an assessment of the underlying paper. Examples, one or two cards per value: claim within evidence: InstructPipe [154].

Support profiles within learning-evidence groups

share of the row's papers carrying each family; 2,761 papers coded in both fields

	what the interface makes reachable				
	Organisation and comparison	Guidance and feedback	Explanation and visibility	User control and expression	Collaboration and perspective
learning evidence					
Learning not measured n = 1,334 papers	73.8% 985	66.9% 893	64.2% 856	52.5% 701	11.4% 152
A proxy measured instead n = 1,039 papers	71.3% 741	74.1% 770	61.5% 639	50.0% 520	14.6% 152
Learning measured n = 258 papers	60.9% 157	82.9% 214	57.4% 148	41.5% 107	14.3% 37
Learning self-reported only n = 130 papers	71.5% 93	84.6% 110	63.8% 83	54.6% 71	20.0% 26
all papers in the matrix n = 2,761 papers	71.6%	72.0%	62.5%	50.7%	13.3%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 53. **Guidance and feedback appear in 83% of capability-measured papers.** Rows: the learning-evidence families, including the papers that measured no capability, with the number of papers in each. Columns: the facet families. Each cell is the share of the row's papers carrying the column family, with the count; colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

C.3 Context and Evidence Profiles

The 2,891 closed-field papers record 2,071 empirical, 691 artifact, 123 methodological, four dataset, and two theoretical contributions. Contribution labels do not identify research methods or target users; study designs and participants are reported in Section 5.2.

In the design space, the domain mix shows no simple annual shift: technical, analytic, and creative professions account for 54.8% in 2019 and 55.5% in 2025, ranging from 53.4% to 61.6% across these years. Annual totals also reflect reading coverage and the incomplete 2026 publication window, so they do not estimate field-wide growth.

What is made reachable, against what the person still does

share of the row's papers in each column; 2,311 papers coded in both fields

what the interface makes reachable	what the person still does			
	The task the tool did not do	New work the tool created	The whole task kept by the person	Final authority kept by design
Organisation and comparison n = 1,639 papers	52.2% 856	29.1% 477	11.4% 187	7.1% 117
Guidance and feedback n = 1,679 papers	49.9% 838	30.1% 506	11.1% 187	8.8% 147
Explanation and visibility n = 1,452 papers	53.2% 772	28.0% 407	10.1% 146	8.7% 126
User control and expression n = 1,170 papers	42.0% 491	35.8% 419	14.2% 166	7.9% 93
Collaboration and perspective n = 311 papers	56.3% 175	19.0% 59	12.9% 40	11.9% 37
all papers in the matrix n = 2,311 papers	51.4%	30.2%	10.1%	8.2%

colour: the row's share against the column's share over all papers; dark is 8 points or more above, pale 8 or more below

Fig. 54. **The profiles share an ordering, with different shares of new work.** The matrix uses the 2,311 systematic papers with complete facet-value and remaining-work assignments. Rows: the facet families, with the number of papers carrying each. Columns: what the person still does. Each cell is the share of the row's papers whose remaining work falls in the column family, with the count. Papers can belong to several mechanism-family rows; each has one remaining-work summary category in this analysis. Colour is the cell's distance from the column's share over all papers in the matrix, given in the last row. The model-assisted grouping is descriptive.

Study design is recorded for 2,071 papers: 691 within-subject designs (33.4%), 328 single-condition usability studies (15.8%), and 119 field deployments (5.7%). The 2,072 entries include one paper with two designs; percentages use papers. Usability studies reveal practical difficulties; deployments can expose work missed by short comparisons.

The participant field also covers 2,071 papers: mixed groups account for 540 (26.1%), target practitioners 499 (24.1%), and crowdworkers 251 (12.1%); 382 (18.4%) are unspecified. Participants need not represent a system's intended users, and missing descriptions do not establish broad applicability.

C.3.1 Most process shifts rest on an author's attribution. Of 8,045 process shifts across 2,891 papers, 6,102 (75.8%) are author-attributed, 1,746 (21.7%) use measured contrasts, and 197 are reviewer inferences. Baselines also differ: 1,940 use a control condition, 869 state none, and the remainder refer to prior practice or an author's assertion. Describing a changed workflow does not establish its effect size.

C.4 Mechanism Families and Remaining Work

The detailed profile complements the control/work comparison in Section 5.2.

D ANNUAL TRENDS AND VENUE RATES

These tests and counts support the comparisons in Section 5 and Section 6. Trends use AI-involved papers with usable codes for each field. Technology follows the AI a paper builds or uses, independently of publication year: generative AI includes hybrids and earlier neural generators; traditional AI/ML includes predictive, discriminative, and symbolic systems. The technology projection covers 3,488 classified papers; 645 remain unassigned. Each annual percentage uses papers with assigned technology and usable coding for that dimension in the same year. Count summaries combine

2019–2026; the annual lines separate the years. Alias disagreements are withheld for the affected field. Adjusted contrasts account for publication year and venue as described below.

D.1 Research-context trends

We model binary context membership with year and venue fixed effects, using venue-clustered standard errors with a finite-sample correction and Student’s t reference distribution. The linear probability coefficient estimates average within-venue change in percentage points per year. Holm adjustment controls the family-wise error rate across the 16 directional tests. Intervals are pointwise 95% confidence intervals. Separate categorical-year tests check temporal variation and departures from a straight trend; these are available with the analysis data. Results describe the coded review frame.

Table 12. Venue-adjusted annual changes in research contexts, in percentage points per year. Primary estimates use 2019–2025; the last column includes available 2026 papers. Sensitivity intervals and adjusted tests are provided online.

Context	Annual change [95% CI]	Adjusted p	With 2026
AI and machine learning practice	-0.59 [-1.17, -0.01]	0.651	-0.64
Data analysis and visualization	-0.35 [-0.90, +0.20]	1.000	-0.46
Software engineering and programming	-0.35 [-0.73, +0.03]	0.881	-0.28
Scientific research practice	+0.35 [-0.03, +0.72]	0.881	+0.39
Information seeking and knowledge work	+0.41 [+0.14, +0.68]	0.065	+0.33
Writing, reading and language	+0.22 [-0.10, +0.55]	1.000	+0.30
Design practice	-0.16 [-0.59, +0.27]	1.000	-0.04
Creative and media production	+0.17 [-0.33, +0.67]	1.000	+0.25
Education and learning	+0.11 [-0.32, +0.54]	1.000	+0.09
Health and clinical care	+0.27 [-0.25, +0.80]	1.000	+0.25
Accessibility and disability	+0.29 [+0.07, +0.52]	0.176	+0.09
Everyday life and personal decisions	-0.28 [-0.60, +0.04]	0.881	-0.08
Online platforms and civic information	-0.09 [-0.48, +0.30]	1.000	-0.03
Physical and embodied activity	-0.18 [-0.61, +0.26]	1.000	-0.11
Business, industry and work	+0.29 [-0.03, +0.60]	0.881	+0.14
Public sector, law and societal systems	-0.04 [-0.20, +0.12]	1.000	-0.11

D.2 Technology comparisons

We compare binary code membership using a linear probability model with technology, categorical publication year, and venue fixed effects. Primary estimates use 2019–2025, and sensitivity models restrict the analysis to 2023–2025 or add partial 2026. Venue-clustered standard errors use a finite-sample correction and a Student t reference with one fewer degree of freedom than the number of venues, and model intervals are pointwise 95% confidence intervals. Holm correction covers all 51 context, activity, family, and category contrasts jointly, with each sensitivity window and venue omission corrected separately. We omit six major venues one at a time: CHI, CSCW, EMNLP, IUI, TVCG, and VIS, each of which supplies at least 5% of the coded papers for workflow, families, or categories in 2019–2025. The choice rests on coding denominators, not outcomes. Weighting checks reweight shared year–venue cells with at least one or at least

Table 13. User control after omitting each major venue and on shared year–venue cells. Gaps are GenAI minus traditional AI/ML in percentage points. Venue-omission gaps are adjusted for year and venue, with pointwise 95% confidence intervals and Holm correction over all 51 outcomes in each fit. Shared-cell gaps are point estimates. Equal-venue gaps also have 95% venue-bootstrap intervals. The one-paper threshold keeps 82.8% of the coded papers, the five-paper threshold 40.8%.

Venue omission	Gap [95% CI]	Holm p
Primary	+13.8 [+8.9, +18.7]	< .001
Omit CHI	+12.8 [+6.9, +18.8]	0.005
Omit CSCW	+13.6 [+8.4, +18.8]	< .001
Omit EMNLP	+14.3 [+9.4, +19.3]	< .001
Omit IUI	+13.9 [+8.5, +19.3]	< .001
Omit TVCG	+12.2 [+7.5, +16.9]	< .001
Omit VIS	+14.5 [+9.7, +19.4]	< .001
Shared-cell weight	At least one per group	At least five per group
Pooled papers	+11.4	+17.3
Within-cell overlap	+12.7	+17.1
Equal cells	+8.5	+16.6
Equal venues	+8.4	+15.1
95% interval	[-2.1, +18.9]	[+9.3, +21.1]
Equal years	+10.3	+18.8

five coded papers per technology by pooled paper count, within-cell overlap, equal cells, equal venues, or equal years. Intervals for equal-venue gaps come from 10,000 resamples of venues.

User control. User control and expression occurs in 634/1,072 GenAI papers (59.1%) and 624/1,381 traditional AI/ML papers (45.2%) in 2019–2025. Omitting each major venue gives gaps from +12.2 to +14.5 points, each with Holm-adjusted support (Table 13). On the 2,031 papers in shared year–venue cells, the gap is +12.7 with overlap weights and +8.4 when venues receive equal weight. Every weighting and cell threshold leaves it positive.

Facets and work codes. The facet and work-code gaps in the main text use the same model on each paper’s latest coding, with a normal reference for intervals. Of the 2,988 technology-classified papers from 2019–2025, 2,532 have facet codes (1,143 GenAI and 1,389 traditional AI/ML), 2,119 have codes for delegated work, and 2,526 have codes for the work people keep. Holm correction runs across the 101 facets coded on at least 40 papers and, separately, across the 24 work codes on at least 30 papers. A gap gets a direction in the text only if it has Holm-adjusted support, keeps its sign under all five weightings at both thresholds, and has an equal-venue interval that excludes zero at one threshold or both (Table 13 and Table 14). Fitted separately for the two models that coded most papers, the control gap is +13.9 (95% CI 8.8 to 19.0; 2,160 papers) and +11.1 (–1.1 to 23.3; 203 papers).

D.3 Temporal change in mechanism families

We distinguish average annual change in family prevalence from annual change in the GenAI–traditional AI/ML difference. Both come from a linear probability model with technology, publication year, their interaction and venue fixed effects. Average change holds the technology mixture fixed at 43.75% GenAI (the original primary cohort’s reference share), separating it from shifts in the composition of the corpus. Positive gap change means movement toward a more positive GenAI difference; it need not enlarge the absolute gap. Venue-clustered intervals use finite-sample correction and a Student t reference. Holm correction covers all ten tests within each window; the 95% confidence intervals are

Table 14. The other gaps given a direction in the text. Gaps are GenAI minus traditional AI/ML in percentage points. Primary gaps are adjusted for year and venue, with 95% confidence intervals. Equal-venue gaps use shared year-venue cells with at least one or at least five papers per technology, with 95% venue-bootstrap intervals. The two thresholds keep 82.7–85.4% and 40.3–46.6% of each coded population. Every gap keeps its sign under all five weightings at both thresholds.

Gap	Primary [95% CI]	Equal venues, papers per group	
		At least one	At least five
Tool generates new artifacts	+16.6 [+12.7, +20.5]	+13.8 [+6.6, +22.3]	+17.7 [+9.3, +29.6]
Tool generates responses and revisions	+9.4 [+5.7, +13.2]	+10.2 [+4.7, +17.1]	+7.6 [+0.8, +14.1]
Tool calculates and models	-11.5 [-15.5, -7.5]	-16.2 [-25.8, -7.2]	-16.2 [-25.6, -7.9]
Tool evaluates and classifies	-9.7 [-14.6, -4.7]	-2.6 [-12.7, +7.9]	-13.2 [-21.5, -2.7]
Tool compares and finds patterns	-6.7 [-9.7, -3.7]	-7.0 [-14.3, -1.1]	-1.2 [-9.5, +8.4]
Generated alternatives to explore	+4.8 [+3.0, +6.6]	+4.3 [+0.9, +8.5]	+1.5 [-1.7, +4.5]
Completion and continuation of content	+3.0 [+1.4, +4.6]	+2.1 [-0.6, +5.3]	+2.1 [+0.6, +3.9]
Person steers the loop	+3.4 [+1.2, +5.6]	+6.7 [+2.4, +11.7]	+0.3 [-5.0, +4.9]
Person improves the system	-5.9 [-7.7, -4.0]	-5.3 [-9.3, -1.8]	-6.4 [-10.9, -2.7]
Organization and comparison (family)	-8.2 [-12.5, -3.8]	-12.2 [-20.4, -4.1]	-8.1 [-17.5, +1.7]
Ranked or prioritized surfacing	-6.5 [-8.3, -4.7]	-3.4 [-7.6, +0.3]	-9.7 [-18.0, -3.7]

Table 15. Mechanism-family trends, 2019–2025: 2,453 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	+1.16 [−0.70, +3.01]	1.000	+1.47 [−0.86, +3.80]	1.000
Organisation and comparison	+0.08 [−1.70, +1.86]	1.000	+0.49 [−2.04, +3.01]	1.000
User control and expression	−1.02 [−2.47, +0.42]	0.961	+0.53 [−1.12, +2.17]	1.000
Guidance and feedback	+1.61 [+0.72, +2.50]	0.007	+1.91 [−0.38, +4.20]	0.692
Collaboration and perspective	+1.13 [+0.53, +1.73]	0.005	+2.15 [−0.02, +4.32]	0.414

Table 16. Mechanism-family trends, 2023–2025: 1,391 coded papers across 39 venues. Changes are percentage points per year; brackets show 95% confidence intervals.

Family	Average annual change [CI]	Holm p	Change in technology difference [CI]	Holm p
Explanation and visibility	−0.34 [−5.73, +5.06]	1.000	+1.44 [−6.17, +9.05]	1.000
Organisation and comparison	−2.86 [−6.65, +0.94]	0.952	−2.86 [−9.68, +3.96]	1.000
User control and expression	−3.41 [−6.55, −0.27]	0.273	−0.41 [−6.16, +5.34]	1.000
Guidance and feedback	+1.92 [−2.69, +6.54]	1.000	+3.49 [−3.80, +10.77]	1.000
Collaboration and perspective	+2.84 [+1.27, +4.41]	0.007	+6.28 [+1.67, +10.89]	0.080

pointwise. These are linear summaries of coded design prevalence, not tests of learning or causal technology effects. Partial 2026 is excluded.

D.4 Annual distributions

Research contexts: volume and annual share

3,307 coded papers with assigned AI technology · categories overlap

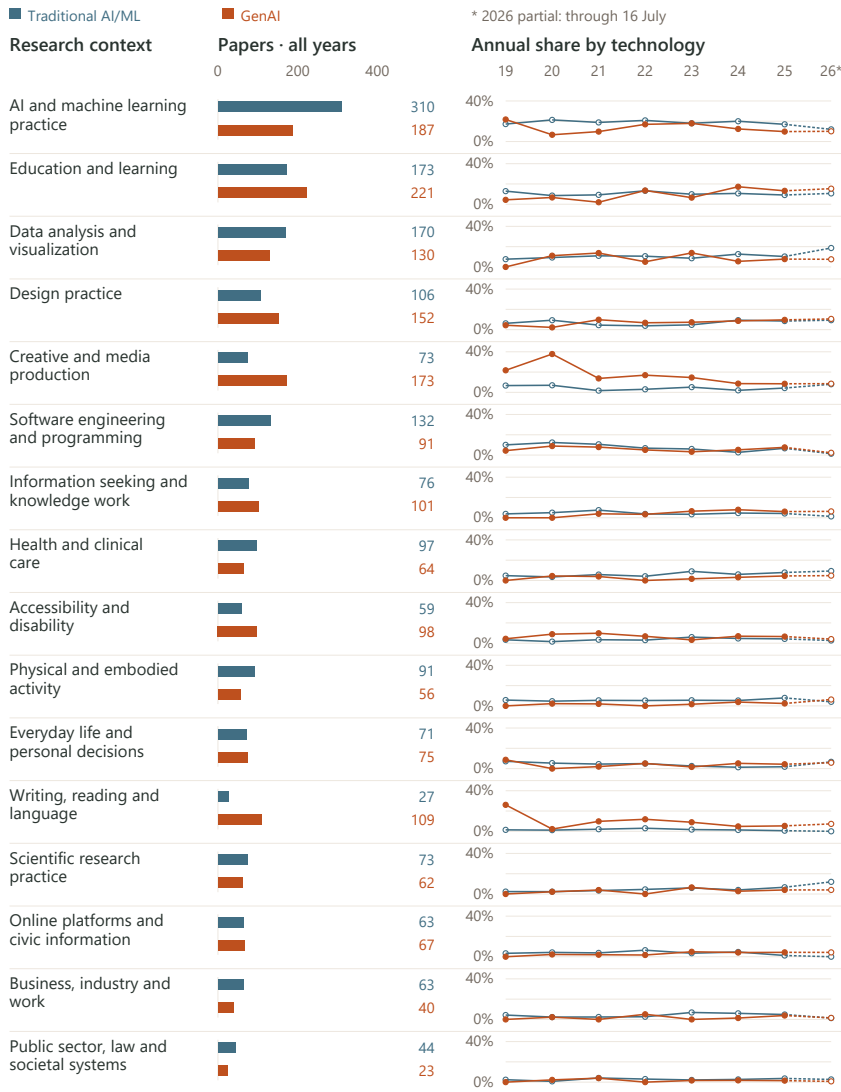
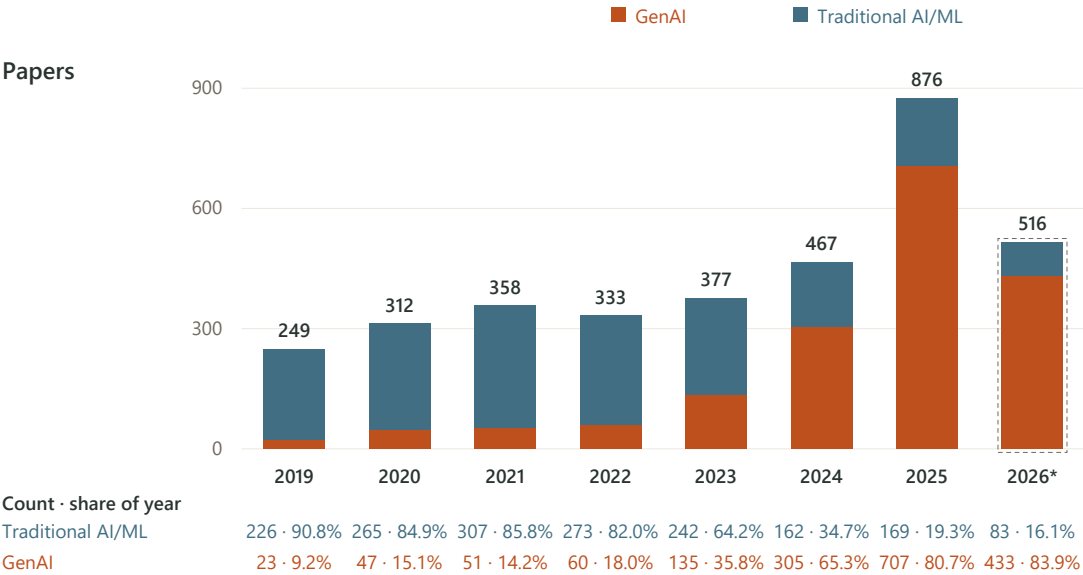


Fig. 55. Research contexts by AI technology. Bars show paper counts across all years; lines show each context's share of its technology group per year, with the coded papers behind those shares below. Contexts overlap. Shares are not adjusted; Table 12 gives the venue-adjusted tests. Only papers with usable context coding and a known technology are included.

AI technologies by year

3,488 classified papers · bar height shows volume; percentages show composition



GenAI includes hybrids and earlier generative models. * 2026 through 16 July; September update pending.

Fig. 56. AI technologies by year. Stacked bar heights encode paper counts, with annual totals above. Rows below give each segment's count and within-year percentage. Classification follows the technology used; 2026 is partial (collected in July). The figure omits 645 papers without a resolved technology classification.

Mechanism families: volume and annual share

2,880 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

■ GenAI

* 2026 partial: through 16 July

Mechanism family

Papers · all years

Annual share by technology

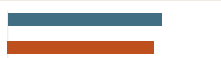
Guidance and feedback



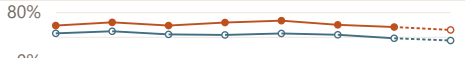
Organisation and comparison



Explanation and visibility



User control and expression



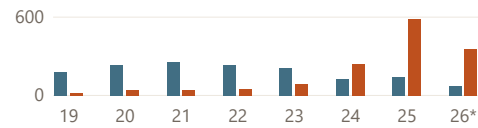
Collaboration and perspective



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.



GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

Fig. 57. Mechanism families by AI technology. Bars encode counts across all years; lines compare annual shares within each technology group. The lower bars show annual coding denominators. Families overlap. Only papers with usable family codes and a known technology are included. Shares are not adjusted; Table 15 gives the venue-adjusted trends.

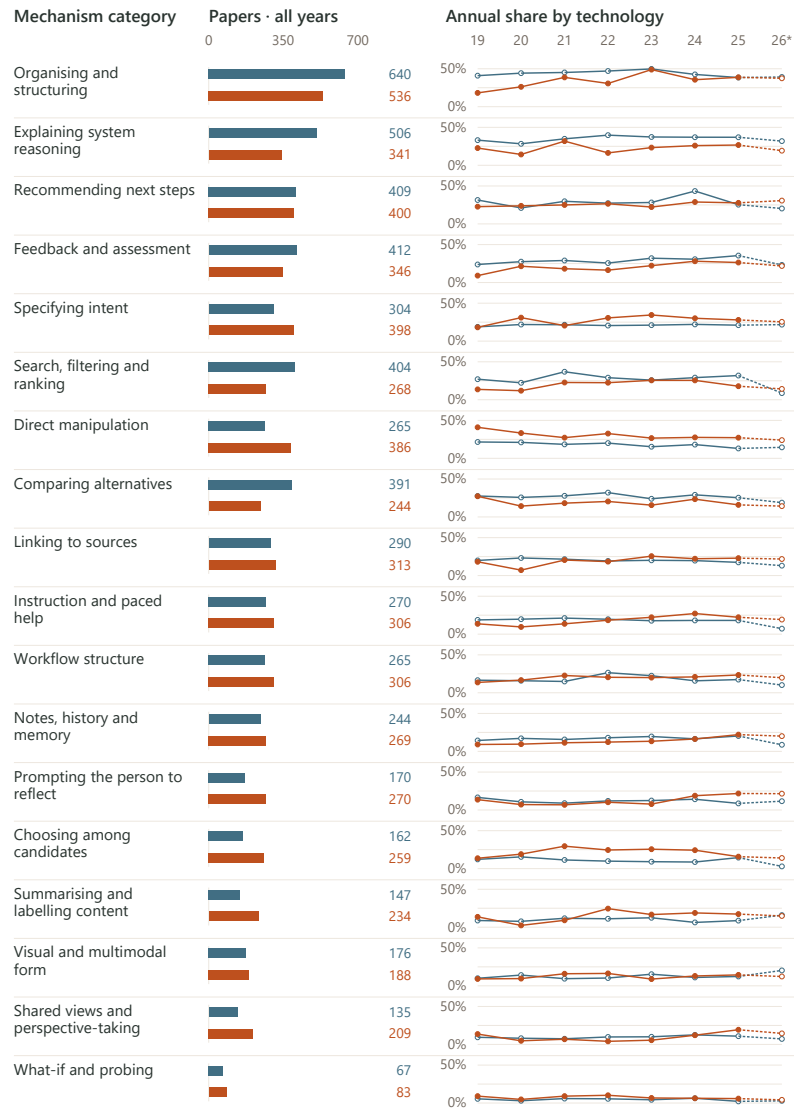
Mechanism categories: volume and annual share

2,874 coded papers with assigned AI technology · categories overlap

■ Traditional AI/ML

■ GenAI

* 2026 partial: through 16 July



Papers behind the annual shares

Each bar counts coded papers available in that technology group and year.

Early GenAI groups are small.

GenAI includes earlier generators and hybrids. These distributions describe designs, not effectiveness.

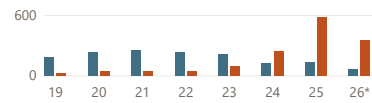


Fig. 58. Mechanism categories by AI technology. Bars encode counts across all years; lines show annual shares within each technology group. Categories overlap, and 2026 covers papers published by 16 July. Shares are not adjusted for year or venue. Only papers with usable category codes and a known technology are included.

D.5 Learning overlays

These counts cover the 3,417 papers with usable category coding, of the 4,133 papers in the analysis at 28 September 2026, including papers without an assigned AI technology. Learning scope means the paper is in the learning review; measured means the paper measured learning explicitly. Learning scope and Measured are counted separately within each category; missing or conflicting labels are omitted from the corresponding column.

Table 17. Papers per mechanism category: all papers, papers in learning scope, and papers that measured learning. Counts are distinct papers; categories overlap.

Category	Papers	Learning scope	Measured
Explaining system reasoning	976	213	97
Linking to sources	683	119	34
Summarising and labelling content	423	107	38
Visual and multimodal form	469	155	55
Organising and structuring	1,382	293	94
Comparing alternatives	731	166	61
Search, filtering and ranking	776	177	71
Notes, history and memory	612	170	68
Specifying intent	820	161	56
Direct manipulation	744	138	56
Choosing among candidates	487	104	37
What-if and probing	171	79	32
Recommending next steps	1,000	330	137
Feedback and assessment	906	318	122
Instruction and paced help	688	347	156
Workflow structure	660	173	70
Prompting the person to reflect	506	242	91
Shared views and perspective-taking	410	163	58

D.6 Venue candidates and current inclusions

Table 18. Venue candidates and AI-involved inclusions in the 16 July 2026 frame, assessed through 28 September 2026. The 37,978 venue-year records represent 37,660 candidate identities. Inclusion (%) is Included divided by Initial candidates. Papers without AI, and papers whose identity or scope is unresolved, are not in Included. Venue memberships can overlap; group and overall totals count each identity once.

Venue	Initial candidates	Included	Inclusion%
HCI & interaction	17,436	2,615	15.0%
56.1% of included corpus			
CHI	7,639	1,152	15.1%
UIST	860	195	22.7%

Venue	Initial candidates	Included	Inclusion%
IUI	687	321	46.7%
DIS	1,471	167	11.4%
CSCW	2,412	212	8.8%
C&C	301	44	14.6%
TOCHI	478	38	7.9%
TiiS	230	95	41.3%
EICS	174	23	13.2%
HAI	214	26	12.1%
ECSCW	205	4	2.0%
GROUP	142	6	4.2%
ASSETS	387	58	15.0%
HRI	630	96	15.2%
IVA	222	39	17.6%
CHIIR	312	40	12.8%
HCOMP	112	28	25.0%
IJHCS	960	71	7.4%
Design & creativity	2,502	102	4.1%
2.2% of included corpus			
ICCC	422	28	6.6%
Design Science	232	15	6.5%
Design Studies	234	1	0.4%
IDETC	1,614	58	3.6%
Visualization	6,336	910	14.4%
19.5% of included corpus			
VIS	1,772	352	19.9%
EuroVis	537	84	15.6%
TVCG	4,073	460	11.3%
VL/HCC	255	65	25.5%
Graphics	2,385	137	5.7%
2.9% of included corpus			
SIGGRAPH	1,673	104	6.2%
SIGGRAPH Asia	712	33	4.6%
Learning & education	2,834	228	8.0%
4.9% of included corpus			
L@S	181	34	18.8%
LAK	655	87	13.3%
AIED	1,168	41	3.5%
IJAIED	347	34	9.8%

Venue	Initial candidates	Included	Inclusion%
EDM	483	32	6.6%
Software engineering 7.6% of included corpus	3,944	354	9.0%
CHASE	249	18	7.2%
ICSE	1,038	129	12.4%
FSE	1,297	88	6.8%
ASE	1,360	119	8.8%
AI & language 6.8% of included corpus	2,223	319	14.3%
ACL	731	133	18.2%
EMNLP	794	146	18.4%
RecSys	698	40	5.7%
Overall (distinct identities)	37,660	4,665	12.4%

E VOCABULARY GROWTH DURING READING

These curves show how each vocabulary grew as cards were read, so a reader can judge how complete the codes are. The vocabulary built from the first 3,295 mechanism cards gained 12 codes across the next 1,754 papers, and each new code was assigned to an existing category and family. All open fields were checked on 821 of those papers; the other batches checked mechanism facets, values, and domains (Figure 59).

E.1 Reading the Curves

We count distinct values after each batch of ten cards, in reading order, and scale each curve to its own final vocabulary. Codes, categories, and families are drawn where those levels exist; categories and families were assigned after coding, so they can reach their final size early in a curve. Reference lists in the lineage cards are open-ended, while their relation options are fixed by the card format. A flat segment means that no new label appeared in those cards, not that unread papers would add none.

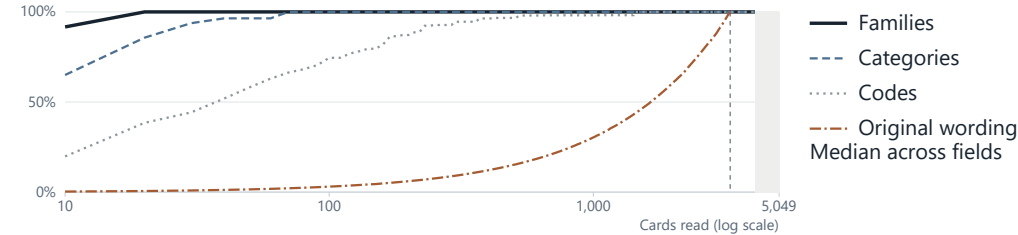
E.2 Stream 1 mechanism cards: Interface and Outcome Fields

These grids show the variation the overview hides. Every panel states its field and final count, and lines are drawn only for the levels present in that field.

Vocabulary growth across three card types

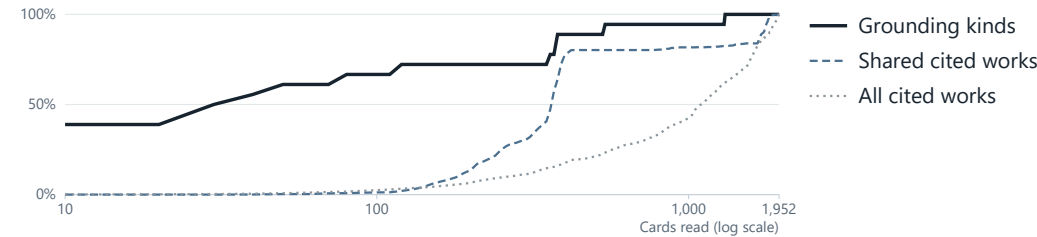
Share of the recorded vocabulary encountered; paper counts use a log scale

Stream 1 · Mechanism cards · 5,049



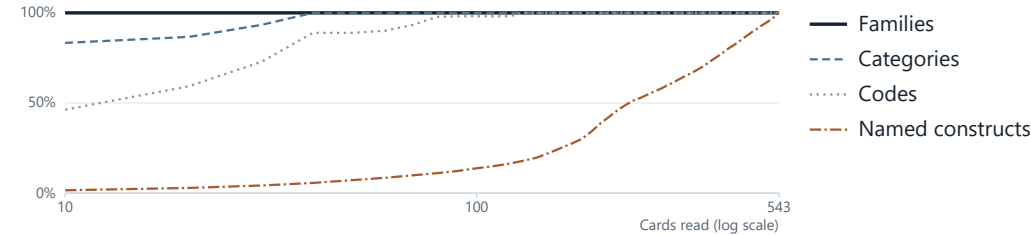
After 3,295: 12 new codes in 1,754 cards; field coverage varies in the shaded region.

Stream 1 · Theory-lineage cards · 1,952



Cited works remain an open vocabulary; grounding kinds are a fixed schema.

Stream 2 · Knowledge-claim cards · 543



Frozen coding covers 542 cards; named constructs cover all 543. Curves stop at their coverage.

Fig. 59. Vocabulary growth across the three card types. Panels count cards in reading order on logarithmic axes: 5,049 mechanism cards, 1,952 theory-lineage cards, and 543 Stream 2 cards. The Stream 1 panels include cards of non-AI comparison papers. Shading marks uneven field coverage, and median curves stop before it.

Codes, categories and families reached, by cards read, every field



Fig. 60. Stream 1, mechanism cards: vocabulary growth in each open field. Each panel separates codes, categories, and families where present, in reading order.

E.3 Stream 1 theory-lineage cards: Cited Sources and Their Roles

The curve covers the 1,952 newly written lineage cards; reused cards appear in the gallery but not in this sequence.

Stream 1 theory-lineage cards: references and relations

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Relations, positions and grounding kinds use fixed options: their flat tails are not evidence of open-code saturation.

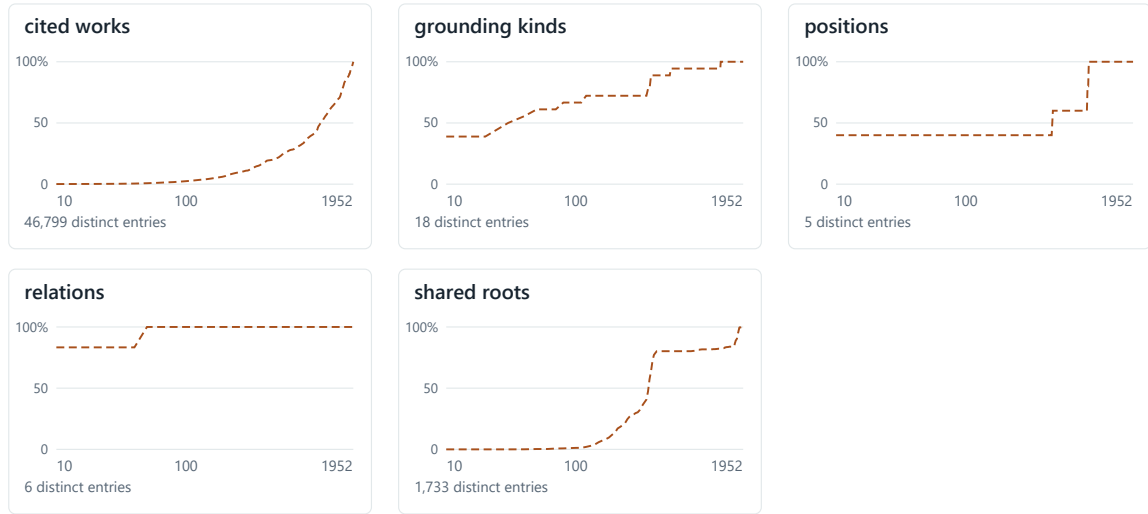


Fig. 61. Stream 1, theory-lineage cards: reference lists keep growing after relation options are exhausted. Fixed options and open-ended lists are separate panels because they close in different ways.

E.4 Stream 2: Claims and Conditions

Stream 2: each coded field

Reading order on a log scale; each curve reaches 100% of its own recorded vocabulary.

Dotted: raw values. Dashed: canonical codes. Solid: categories. Thick: families, where present.

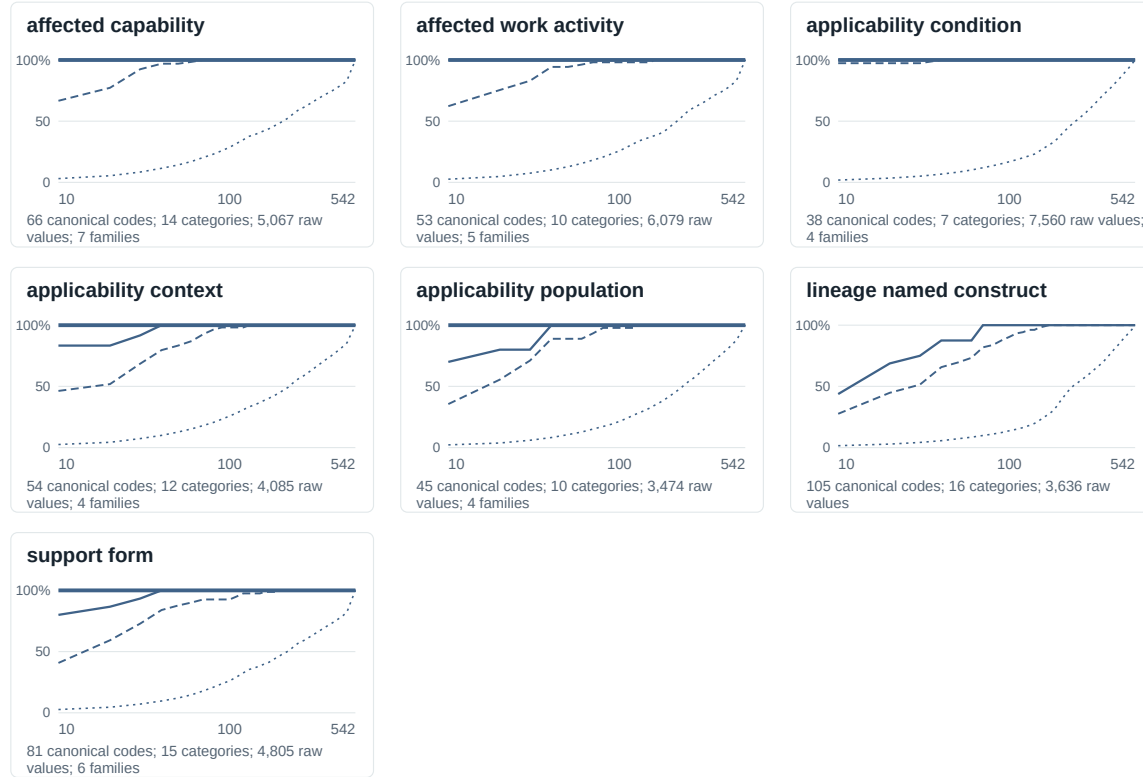


Fig. 62. **Stream 2: vocabulary growth differs by coded field.** Codes, categories, and families are drawn only where that level exists.

Stream 2's seven open fields cover 542 cards and 35,699 observations. Of the 449 codes, 448 are used, and no new code entered in the last 16 coding batches. A flat curve can still hide a missing code: no population code names a studied unit that is or includes a machine, and 18 assignments flag this while using codes for people or artifacts.

E.5 New Codes Within the First Coding

Within the first coding of each open field, the last new code appeared between 2.5% and 48.7% of the way through the fields whose coding order was kept (Figure 63). Three fields still have low-confidence assignments for more than a quarter of their observations, so a vocabulary can stop growing and remain too coarse.

Where did the recorded vocabulary stop expanding?

Colour ends at the last new code. The pale remainder reuses codes already encountered.

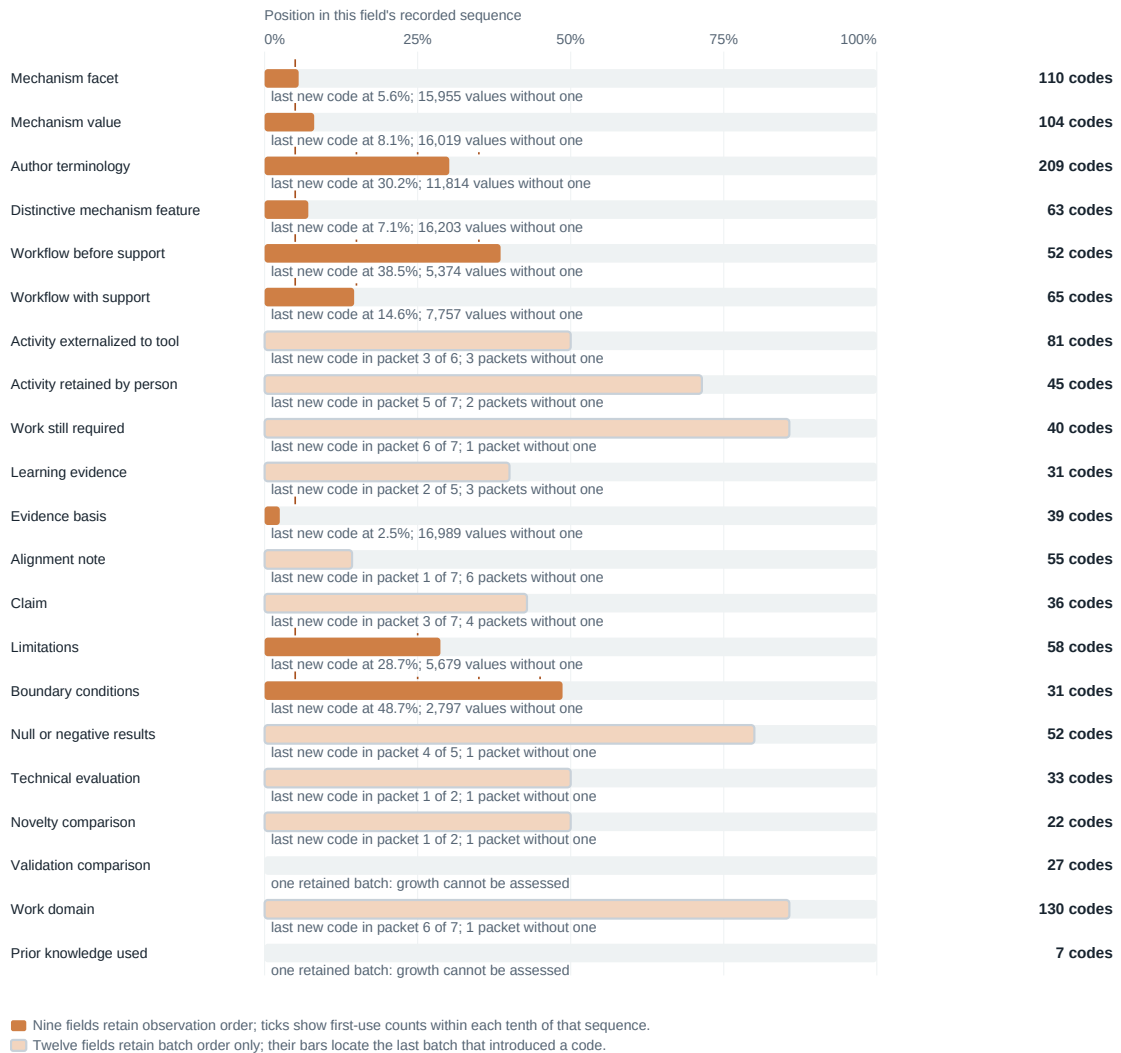


Fig. 63. **New-code creation within each field of the first coding.** Bars locate the last new code in the recorded coding sequence; ticks show new codes by tenth of the sequence. Where only batch order survives, the figure reports batch resolution.